

Dynamic-Reference Cumulative Prospect Policy Gradients: Debiased Estimation, Minimax Limits, and Online Portfolio Tracking

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Abstract

Cumulative prospect theory (CPT) makes portfolio evaluation reference dependent, yet first-order CPT policy optimization is usually formulated around a fixed reference. We study the harder setting in which realized wealth updates the reference state, so the learning target is simultaneously nonlinear, path dependent, and nonstationary. We introduce Dynamic-Reference Cumulative Prospect Policy Gradient (DRCPT-PG), combining a factor-parametrized Dirichlet portfolio policy with asymmetric reference adaptation, an exactly unbiased frozen-target CPT gradient estimator, and normalized projected online tracking. The estimator has finite expected trajectory cost and variance, admits studentized asymptotic inference, and attains the fixed-dimensional minimax trajectory-budget rate; the online analysis controls the projected residual and dynamic local regret under sublinear target variation. In controlled experiments, continued updating reduces cumulative squared projected residual from 1.427 to 0.684 relative to a frozen closed-loop control. The external-validity layers preserve the same separation of claims: in a time-ordered 300-stock A-share replay, DRCPT-PG has the highest mean Sharpe ratio and the lowest mean drawdown and turnover among the learned policies; on 71,854 held-out investor decisions, the asymmetric reference lowers log loss by 0.0249 relative to no reference under paired investor-cluster bootstrap; and in a four-type calibrated interaction replay it gives the highest adoption-weighted satisfaction among the learned advisers in every investor type. Together, the results show how endogenous reference formation can be treated as a first-order statistical and online-optimization problem while retaining market and behavioral interpretability.

1 Introduction

Portfolio advice is inherently sequential and path dependent. An investor who reaches the same current wealth after a rally and drawdown may evaluate the next risky allocation very differently from an investor who reaches that wealth after a loss and recovery. In human-facing financial systems, this distinction matters because a policy that is attractive under expected return can be rejected once recent gains, losses, and reference levels are taken into account. Cumulative prospect theory (CPT) provides a canonical description of such behavior through reference dependence, asymmetric valuation of gains and losses, and nonlinear probability weighting [1, 2]. These features make CPT particularly relevant to sequential portfolio choice, but they also turn the learning objective into a nonlinear functional of the entire trajectory-outcome distribution rather than an additive one-period reward.

A growing literature has brought this distributional objective into reinforcement learning. Early work formulated prediction and control under CPT and established stochastic optimization procedures for prospect-theoretic objectives [3, 4]. Most closely related to this paper, Lepel and Barakat [5]—the current v5 version of the core CPT policy-gradient reference—derive a first-order policy-gradient theorem for finite-horizon CPT-RL, construct an order-statistics Monte Carlo estimator, and prove convergence and sample-complexity guarantees for first-order stationarity.

This establishes a principled foundation for CPT policy optimization. Its reference point, however, enters as a fixed element of the CPT criterion rather than an endogenous state that is updated by the realized wealth path.

Behavioral evidence and recent portfolio models point in the opposite direction. Reference points adapt after realized gains and losses, with systematic asymmetry in the speed of adjustment [6, 7]; adaptive and path-dependent references can materially change realization and portfolio decisions [8–10]. The simplest example already changes the optimization problem. Starting from normalized wealth 1, consider a rise to 1.4 followed by a fall to 1.2, versus a fall to 0.8 followed by a recovery to 1.2. The terminal wealth is identical, but an upward-adjusted reference can place the first path in the loss domain while a lower reference places the second in the gain domain. Portfolio-level evidence that gain–loss framing affects the disposition effect reinforces the importance of this state dependence [11], while recent Chinese retail-investor evidence shows that disposition behavior also varies with information and investor characteristics [12].

Once the reference point is made endogenous, two difficulties that are separate under a fixed target become coupled. First, the CPT gradient depends on the derivative of a probability-weighting function evaluated at an unknown outcome distribution. Finite-rank plug-in evaluation therefore introduces a nonlinear estimation problem before the policy can be updated. Second, the target itself moves: market conditions generate exogenous variation, while the policy changes wealth, wealth changes the reference point, and the reference point changes the next CPT objective. Convergence to a single stationary point is no longer the appropriate criterion. The relevant question is whether a feasible first-order policy can track a sequence of moving stationary conditions, and how its tracking error decomposes into target variation and statistical estimation error.

This paper develops *Dynamic-Reference Cumulative Prospect Policy Gradient* (DRCPT-PG) for this setting. The portfolio policy is a factor-parametrized Dirichlet distribution, which produces long-only simplex-valued allocations while keeping the parameter dimension tied to interpretable characteristics rather than the number of stocks. The reference state follows an asymmetric update after realized wealth changes. For learning, a centered leave-one-out score statistic is combined with an independent, untruncated randomized full/half correction. The resulting gradient output is exactly unbiased for the frozen regularized CPT target, has finite variance and finite expected trajectory cost, admits a studentized central limit theorem, and attains the fixed-dimensional minimax trajectory-budget rate. The online layer then uses a normalized projected update and evaluates tracking through the projected residual and dynamic local regret (DLR).

The analysis is organized around three linked questions. The first is statistical: can the nonlinear CPT policy gradient be estimated without finite-sample bias while retaining usable variance, inference, and budget guarantees? The second is dynamic: under what variation and simulation-error conditions does online updating keep the projected first-order residual small as the market and reference state evolve? The third is behavioral and empirical: does dynamic reference formation produce the path dependence and trading asymmetries predicted by behavioral finance, and do the resulting distinctions remain visible when the model is taken from controlled simulations to real A-share portfolios, held-out investor behavior, and heterogeneous retail-agent interactions? This separation lets each layer of evidence test a distinct part of the argument rather than using portfolio returns as a proxy for estimator correctness or using behavioral fit as a proxy for optimization quality.

The theoretical results answer the first two questions. We derive the regularized CPT score-gradient identity; prove the finite-sample bias order of the centered leave-one-out base; construct an exactly unbiased estimator with a finite M^{-1} variance bound, asymptotic normality, and a minimax-optimal $\Theta(B^{-1})$ fixed-dimensional trajectory-budget rate; and establish a bound on the average expected squared projected residual for the online algorithm. Dynamic local regret is

bounded by the cumulative squared projected residual, and a local error-bound condition converts the same residual control into distance to the moving stationary set. When objective variation is sublinear and the sampling and simulation errors vanish in Cesàro average, the expected average DLR vanishes.

The empirical program mirrors this logic. Path experiments first isolate the role of reference history in portfolio ranking. Fixed-target experiments then evaluate the estimator independently of online state propagation, continuous slow-variation experiments test online tracking, and dense sensitivity sweeps map the region over which the tracking advantage persists. The external-validity layers then move from controlled objects to market and behavioral outcomes. In the time-ordered A-share replay, DRCPT-PG has the highest mean Sharpe ratio (0.856), the lowest mean maximum drawdown (11.67%), and the lowest mean turnover (10.87) among the learned policies. On 71,854 held-out investor decisions, the asymmetric reference representation gives the best log loss, Brier score, AUROC, and AUPRC among the four matched predictive specifications; the paired bootstrap log-loss improvement over no reference is 0.0249 with a 95% interval [0.0214, 0.0287]. Finally, replaying the E5 recommendations through four investor types calibrated from real behavior yields the highest adoption-weighted satisfaction for DRCPT-PG among learned advisers in every type. This progression connects theorem-level objects—gradient error, projected residual, and DLR—to economic, behavioral, and interaction outcomes while keeping their estimands distinct.

Contributions.

1. **Endogenous-reference CPT policy optimization.** We formulate a finite-horizon portfolio objective in which the psychological reference point is part of the evolving state rather than a fixed hyperparameter. The model yields explicit path dependence and a decomposition of objective variation into non-reference and reference components; fixed-reference CPT-PG is recovered as a special case.
2. **Debiased gradient estimation with statistical guarantees.** We establish the regularized CPT score-gradient identity and construct a centered leave-one-out estimator with an independent randomized raw correction. The estimator is exactly unbiased, has finite expected cost and finite variance, obeys a multivariate central limit theorem, and attains the fixed-dimensional minimax rate under the same expected complete-trajectory budget.
3. **Online first-order tracking for a moving nonconvex target.** We analyze a normalized projected update through the projected residual. The resulting bound separates objective variation, replication error, and frozen-simulator error; it implies a DLR bound and, under an optional local error-bound condition, tracking of the moving stationary set.
4. **Layered external validity from market replay to behavior and interaction.** Beyond the theory-aligned E1–E4 sequence, a fixed-universe 300-stock A-share replay shows a favorable risk-adjusted and turnover profile among learned policies; a matched held-out behavior test shows stable incremental predictive value from asymmetric reference representation; and a calibrated four-type interaction replay preserves the strongest adoption-weighted satisfaction among learned advisers across investor types. These layers extend the same reference mechanism without using portfolio return, predictive fit, or simulated satisfaction as substitutes for the estimator and tracking diagnostics.

Organization. [Section 2](#) positions the paper relative to CPT optimization, adaptive-reference finance, nonstationary nonconvex learning, statistical debiasing, and human-facing financial agents.

Section 3 defines the dynamic-reference portfolio objective, projected tracking criterion, and the assumptions needed for the statistical and online analyses. Section 4 presents the debiased gradient estimator and DRCPT-PG update. Section 5 gives the estimator, minimax, projected-residual, DLR, and stationary-set results. Section 6 organizes the mechanism, estimator, tracking, robustness, market, behavioral, and heterogeneous-agent evidence. Section 7 synthesizes the implications and scope; proofs are collected in the appendix.

2 Related Work and Positioning

CPT optimization and reinforcement learning. Prospect theory and CPT replace expected-utility evaluation by a reference-dependent value transformation and nonlinear probability weighting [1, 2]. In sequential decision making, Prashanth et al. [3] showed that a CPT criterion must be treated as a functional of the return distribution rather than a transformed immediate reward, and Jie et al. [4] developed stochastic-optimization machinery for this class of objectives. The closest reference is Lepel and Barakat [5], whose v5/TMLR formulation derives a first-order CPT policy-gradient theorem, an order-statistics Monte Carlo estimator, and convergence and sample-complexity guarantees for approximate first-order stationarity. In parallel, recent optimization work has developed finite-scenario and portfolio-specific algorithms for static CPT objectives, including ADMM-based decision optimization [13], convex-analytic portfolio procedures [14], and a recent behavioral portfolio formulation that combines CPT with a flexible investment horizon [15]. Our work keeps the first-order CPT policy-gradient perspective but changes two structural elements: the reference becomes an endogenous state generated by realized wealth, and the learning objective therefore becomes a moving, path-dependent sequence. This requires both exact finite-sample debiasing of the frozen-target gradient and a tracking analysis for the resulting moving objective.

Adaptive references, path dependence, and investor behavior. Reference adaptation has long been documented in security-trading experiments, including asymmetric updating after equal-sized gains and losses [6, 7]. Mathematical-finance models then showed that adaptive references can amplify realization behavior [8], while path-dependent reference points make current decisions depend on the route by which wealth arrived at its present level [9]. More recent portfolio analysis embeds a dynamic reference directly in multi-period loss-averse allocation and finds materially different policies and disposition-effect predictions [10]. Complementary evidence shows that the disposition effect depends on portfolio-wide gain/loss frames [11], investor horizon and experience [16], and the information environment of Chinese retail investors [12]. These studies establish that the relevant behavioral state is neither fixed nor purely asset-local. They do not, however, provide an online CPT policy-gradient procedure whose parameter update and first-order tracking guarantees explicitly accommodate reference drift. That is the point at which the behavioral-reference literature and the optimization problem studied here meet.

Nonstationary and nonconvex online optimization. For nonconvex objectives, global comparator regret is generally too strong to characterize first-order learning. Local-regret criteria instead measure whether recent gradients or gradient mappings are small [17]. Aydoore et al. [18] introduced dynamic local regret for nonconvex forecasting with time-smoothed gradients, and Hallak et al. [19] extended local-regret ideas to stochastic constrained problems through a proximal-gradient mapping. In nonstationary RL, variation budgets similarly quantify when a changing environment remains learnable [20]. Recent projected-gradient analysis sharpens first-order stationarity guarantees for constrained nonconvex and stochastic smooth optimization [21]. Our setting combines these

ingredients but introduces a distinct source of variation: part of the objective drift is generated by the learner’s own realized wealth and reference dynamics. We therefore work with a normalized projected residual, derive DLR from that residual, and isolate objective variation and simulator error in the online bound.

Debiasing, inference, and information limits. The statistical difficulty in a CPT gradient is generated by a nonlinear functional of an unknown survival distribution. The order-statistics approach of Lepel and Barakat [5] provides a practical first-order estimator and consistency guarantees. Our estimator targets a different finite-sample property: exact unbiasedness for the regularized frozen-target gradient under a complete-trajectory oracle. Its independent randomized correction follows the general principle of unbiased randomized telescoping [22], while the leave-one-out expansion uses uniform empirical-distribution control of the type underlying the Dvoretzky–Kiefer–Wolfowitz inequality [23]. The central-limit and minimax layers connect the estimator to classical asymptotic and statistical-decision theory [24, 25]. The resulting contribution is therefore not merely a numerical bias correction: it specifies the mean, variance, limiting distribution, expected trajectory cost, and fixed-dimensional information rate of the gradient oracle used by the online algorithm.

Portfolio learning and A-share evaluation. The portfolio policy combines factor information with a simplex-valued action. Dirichlet factor policies provide a natural precedent for reinforced portfolio allocation because they generate valid long-only weights while linking concentration parameters to firm characteristics [26]. For the empirical market layer, recent work emphasizes both the time variation of financial-RL environments [27] and the importance of market-specific factor structure in Chinese A-shares [28]. These considerations motivate separating the controlled semi-synthetic environment used for theorem-aligned E2–E4 from a chronological real-market test. Our E5 implements that separation on a fixed 300-stock HS300 universe with strictly point-in-time factors, common transaction costs, and the same stochastic Dirichlet execution rule used by the learner. The resulting return, drawdown, turnover, and cost outcomes are therefore an external-validity layer for the portfolio mechanism rather than a surrogate test of the DLR theorem.

Human-facing financial agents and heterogeneous risk preferences. Recent AI research increasingly treats behavioral preferences as part of the system objective rather than as evaluation noise. Prospect-theoretic optimization has been used to construct human-aware alignment losses for language models [29], and recent work directly evaluates whether LLMs represent persona-specific economic risk preferences [30]. In multi-agent learning, Lalmohammed et al. [31] embed CPT transformations into heterogeneous agents and show that risk profiles can materially change strategic behavior. Financial-advice studies add an interaction layer: personalized generative-AI advisors can influence investment choices, but preference elicitation, advice quality, satisfaction, and trust need not move together [32], while investor resistance to robo-advisors is shaped by both functional and psychological barriers [33]. Our behavioral design separates two empirical questions that this literature suggests should not be conflated. E6 holds investor decisions, controls, chronological split, and predictive model fixed while replacing only the reference representation, so its evidence concerns held-out behavioral alignment. E7 then calibrates heterogeneous investor types from training-period behavior and replays the E5 recommendations with adoption decided before episode returns are observed, separating recommendation uptake from satisfaction after adoption. This connects dynamic-reference portfolio learning to human-facing interaction without treating simulated satisfaction as observed behavior.

3 Problem Setup

3.1 Nonstationary episodic portfolio process

Episodes are indexed by $k = 1, \dots, K$, begin at dates t_k , and have a fixed horizon $h \geq 1$. For consecutive episodes, $t_{k+1} = t_k + h$. At calendar time $u \in \{t_k, \dots, t_k + h - 1\}$,

$$s_u = (m_u, X_u, r_u, \omega_{u-1}), \quad \omega_u \in \Delta_N := \left\{ \omega \in \mathbb{R}_+^{N+1} : \sum_{i=0}^N \omega_i = 1 \right\}, \quad (1)$$

where asset 0 is cash, $X_u > 0$ is normalized wealth, and $r_u \geq 0$ is the reference wealth. For return vector R_{u+1} with $R_{0,u+1} = 0$ and proportional transaction-cost coefficient $c_{tc} \geq 0$,

$$X_{u+1} = X_u \left(1 + \omega_u^\top R_{u+1} - c_{tc} \|\omega_u - \omega_{u-1}\|_1 \right), \quad X_1 = 1. \quad (2)$$

The conditional market transition law may vary with calendar time. Conditional on the observed history and action, it has no explicit dependence on the policy parameter.

3.2 Dynamic reference and terminal relative outcome

For fixed $\eta_+, \eta_- \in [0, 1]$,

$$r_{u+1} = \Phi(r_u, X_{u+1}) = \begin{cases} (1 - \eta_+)r_u + \eta_+ X_{u+1}, & X_{u+1} \geq r_u, \\ (1 - \eta_-)r_u + \eta_- X_{u+1}, & X_{u+1} < r_u. \end{cases} \quad (3)$$

Let $\mathcal{F}_k$ be the information available at the start of episode k , and set

$$r_k := r_{t_k}, \quad z_k := (m_{t_k}, X_{t_k}, \omega_{t_k-1}). \quad (4)$$

All Monte Carlo learning trajectories for episode k are generated conditionally on the frozen context $(\mathcal{F}_k, z_k, r_k)$. Each trajectory τ carries its own wealth and reference recursions and produces

$$Y_k(\tau) = X_{t_k+h}(\tau) - r_{t_k+h}(\tau). \quad (5)$$

3.3 Dirichlet portfolio policy

For asset $i = 0, \dots, N$, let $q_{i,u} \in \mathbb{R}^d$ be a bounded feature vector. For $\theta \in \Theta \subset \mathbb{R}^d$ define

$$a_{i,u}(\theta) = \exp(\theta^\top q_{i,u}), \quad \tilde{\omega}_u \sim \pi_\theta(\cdot \mid s_u) = \text{Dirichlet}(a_{0,u}(\theta), \dots, a_{N,u}(\theta)). \quad (6)$$

A parameter-independent feasibility map $\mathbf{M}_u : \Delta_N \rightarrow \Delta_N$ and a fixed trade fraction $\chi \in (0, 1]$ generate the executed portfolio

$$\omega_u = (1 - \chi)\omega_{u-1} + \chi \mathbf{M}_u(\tilde{\omega}_u). \quad (7)$$

The realized path and all learning trajectories use the same stochastic actor (6) and the same map (7). The augmented trajectory records $\tilde{\omega}_u$, so its score is

$$G_{k,\theta}(\tau) = \sum_{u=t_k}^{t_k+h-1} \sum_{i=0}^N a_{i,u} q_{i,u} \left\{ \log \tilde{\omega}_{i,u} - \psi(a_{i,u}) + \psi \left(\sum_{j=0}^N a_{j,u} \right) \right\}, \quad (8)$$

where ψ is the digamma function.

3.4 Regularized CPT objective

For $x \geq 0$ let

$$u_{\alpha, \varepsilon_v}(x) = (x^2 + \varepsilon_v^2)^{\alpha/2} - \varepsilon_v^\alpha, \quad v_+(y) = u_{\alpha_+, \varepsilon_v}(y^+), \quad v_-(y) = \lambda u_{\alpha_-, \varepsilon_v}(y^-), \quad (9)$$

with $0 < \alpha_+, \alpha_- \leq 1$, $\lambda > 1$, and $\varepsilon_v > 0$. Let $w_\pm$ be strictly increasing probability-weighting functions and, for fixed $\varepsilon \in (0, 1/2)$, define the endpoint-normalized weights

$$w_\pm^\varepsilon(p) = \frac{w_\pm(\varepsilon + (1 - 2\varepsilon)p) - w_\pm(\varepsilon)}{w_\pm(1 - \varepsilon) - w_\pm(\varepsilon)}, \quad 0 \leq p \leq 1. \quad (10)$$

Thus $w_\pm^\varepsilon(0) = 0$ and $w_\pm^\varepsilon(1) = 1$.

For a frozen episode environment/model P and context (z, r) , let P_θ denote the induced augmented trajectory law under π_θ . Write

$$S_{P, z, r, \theta}^\pm(y) = P_\theta\{v_\pm(Y) > y\}.$$

On the bounded analysis domain, $v_\pm(Y) \in [0, B_v]$. Define

$$J(P, z, r; \theta) = \int_0^{B_v} w_+^\varepsilon\left(S_{P, z, r, \theta}^+(y)\right) dy - \int_0^{B_v} w_-^\varepsilon\left(S_{P, z, r, \theta}^-(y)\right) dy. \quad (11)$$

Let P_k denote the true conditional episode- k model and $\tilde{P}_k$ its frozen simulator model. Both are fixed when differentiating with respect to θ ; their induced trajectory laws depend on θ through the policy. Then

$$J_k(\theta, r_k) := J(P_k, z_k, r_k; \theta), \quad \tilde{J}_k(\theta, r_k) := J(\tilde{P}_k, z_k, r_k; \theta), \quad (12)$$

with gradients

$$g_k(\theta) := \nabla_\theta J_k(\theta, r_k), \quad \tilde{g}_k(\theta) := \nabla_\theta \tilde{J}_k(\theta, r_k). \quad (13)$$

3.5 Projected residual and dynamic local regret

Fix a normalization floor $\vartheta > 0$ and a step size $\gamma > 0$. Define

$$\mathbf{d}(g) = \frac{g}{\max\{\|g\|_2, \vartheta\}}, \quad (14)$$

and the population projected residual

$$Q_k(\theta) = \frac{\Pi_\Theta\{\theta + \gamma \mathbf{d}(g_k(\theta))\} - \theta}{\gamma}. \quad (15)$$

The constrained first-order stationary set is

$$\mathbf{S}_k = \{\theta \in \Theta : Q_k(\theta) = 0\} = \{\theta \in \Theta : \langle g_k(\theta), u \rangle - \theta \leq 0 \ \forall u \in \Theta\}. \quad (16)$$

Because J_k is continuously differentiable on compact convex Θ , each J_k attains a maximizer satisfying (16); continuity of Q_k then makes $\mathbf{S}_k$ a nonempty compact set. At the realized iterate, let

$$\zeta_k := \|Q_k(\theta_k)\|_2^2, \quad \mathcal{E}_K := \frac{1}{K} \sum_{k=1}^K \mathbb{E} \zeta_k. \quad (17)$$

For window length $w \geq 1$ and discount $\rho \in (0, 1]$, set

$$W_{\rho,w} := \sum_{j=0}^{w-1} \rho^j, \quad \bar{Q}_k := W_{\rho,w}^{-1} \sum_{j=0}^{w-1} \rho^j Q_{k-j}(\theta_{k-j}), \quad (18)$$

with $Q_t(\theta_t) := 0$ for $t \leq 0$. The dynamic local regret (DLR) is

$$\mathcal{R}_{\rho,w}(K) := \sum_{k=1}^K \|\bar{Q}_k\|_2^2. \quad (19)$$

Here Q_k is the projected residual, ζ_k is the squared projected residual, and $\mathcal{E}_K$ is the average expected squared projected residual. Each projected residual is evaluated at its contemporaneous iterate before the residual vectors are exponentially averaged over the recent window. This is the dynamic local-regret object used to evaluate online adaptation below.

3.6 Assumptions

3.6.1 Standing statistical assumptions

Assumption 3.1 (Bounded analysis domain and likelihood regularity). *The parameter set $\Theta \subset \mathbb{R}^d$ is nonempty, compact, and convex, and is contained in an open set on which the policy is defined. The horizon h is finite and $N \geq 1$. There exist finite constants Q_f, B_X, B_r, B_v such that, on the analyzed domain,*

$$\sup_{i,u} \|q_{i,u}\|_2 \leq Q_f, \quad 0 < X_u \leq B_X, \quad 0 \leq r_u \leq B_r, \quad 0 \leq v_{\pm}(Y_k) \leq B_v. \quad (20)$$

Conditional on the observed history and action, the market transition kernel and the map $\mathbf{M}_u$ do not depend explicitly on θ . The latent Dirichlet proposals are retained in the augmented trajectory. Hence all explicit θ -dependence of the conditional trajectory law enters through the Dirichlet policy density in (6). The bounds and parameter-independence conditions above are required uniformly for every analyzed true episode model P_k and frozen simulator model $\tilde{P}_k$, conditional on the episode-start information and context. In particular, the frozen simulator is held fixed as θ varies.

Assumption 3.2 (Fixed regularized CPT). *The preference parameters in (9)–(10) are fixed. The functions $w_{\pm}^{\varepsilon}$ are strictly increasing and belong to $C^3([0, 1])$, with*

$$C_j := \max_{\sigma \in \{+, -\}} \sup_{p \in [0, 1]} |(w_{\sigma}^{\varepsilon})^{(j)}(p)| < \infty, \quad j = 1, 2, 3. \quad (21)$$

Under Assumptions 3.1–3.2, Appendix A derives finite uniform score moments and constants $B_g, L < \infty$ such that

$$\sup_{k, \theta \in \Theta} \|g_k(\theta)\|_2 \leq B_g, \quad \|g_k(\theta) - g_k(\theta')\|_2 \leq L \|\theta - \theta'\|_2. \quad (22)$$

3.6.2 Online variation and optional geometric conditions

Assumption 3.3 (Frozen-simulator error and objective variation). *For the queried parameter θ_k in episode k ,*

$$\|\tilde{g}_k(\theta_k) - g_k(\theta_k)\|_2 \leq \beta_k, \quad \mathbb{E}\beta_k^2 < \infty, \quad (23)$$

where β_k is $\mathcal{F}_k$ -measurable. Define

$$\mathcal{V}_K := \sum_{k=1}^{K-1} \sup_{\theta \in \Theta} |J_{k+1}(\theta, r_{k+1}) - J_k(\theta, r_k)|. \quad (24)$$

For the finite horizon under study, $\mathbb{E}\mathcal{V}_K < \infty$.

Assumption 3.4 (Stationary-set error bound; optional). *There exists $\kappa < \infty$ such that, uniformly over the analyzed episodes,*

$$\text{dist}(\theta, \mathcal{S}_k) \leq \kappa \|Q_k(\theta)\|_2, \quad \theta \in \Theta. \quad (25)$$

Assumption 3.5 (Reference-isolated coupling; optional). *When two episode-start references are compared, the market law, non-reference start state, and policy parameter are held fixed; the policy features and market transition kernel do not depend directly on the reference value. The two trajectories can therefore be coupled with common policy and market random variables, yielding the same wealth and portfolio paths.*

3.6.3 Fixed-target minimax experiment

Assumption 3.6 (Exact-trajectory minimax experiment). *Fix d , an interior point $\theta_0 = 0 \in \Theta$, finite upper bounds on h and N , and the common constants in Assumptions 3.1–3.2. Let $\mathfrak{C}$ be a class of conditional portfolio models satisfying these bounds and containing the Bernoulli-return submodel constructed in Appendix E; assume $(w_+^e)'(0) > 0$.*

An estimator may adaptively query any $\theta \in \Theta$. Each oracle call returns one fresh complete augmented trajectory from the queried policy under the unknown $P \in \mathfrak{C}$. The procedure may randomize and stop at a data-dependent call count T_{call} , subject to

$$\sup_{P \in \mathfrak{C}} \mathbb{E}_P T_{\text{call}} \leq B. \quad (26)$$

No additional observation of the unknown market law is available. The target is $g_P = \nabla J_P(\theta_0)$, while d and all domain constants remain fixed as $B \rightarrow \infty$.

3.6.4 Assumption roles and scope

Table 1: Role, verification route, and scope fallback of the assumptions. The fallback column states which conclusion is retained when the corresponding optional or asymptotic condition is unavailable.

Assumption	Role	Verification route	If unavailable
3.1	score identity, moments, smoothness	parameter box; bounded factors; Appendix A	re-establish the required likelihood moments on the enlarged domain
3.2	Taylor expansion and fixed-target debiasing	analytic derivatives of the fixed regularized weights	change the regularization or prove a replacement remainder bound
3.3	true-target online residual bound	simulator validation for β_k ; direct objective-difference bound for $\mathcal{V}_K$	finite-horizon theorem remains with the observed error terms; vanishing tracking is not asserted
3.4	residual $\Rightarrow$ distance to $\mathbf{S}_k$	objective-specific geometry or a proved local error bound	retain residual and DLR conclusions only
3.5	explicit reference-only sensitivity decomposition	verify that features and market kernel do not use r directly	retain the general variation budget $\mathcal{V}_K$ without the reference-only split
3.6	fixed- d minimax information rate	check that the information source and budget match the oracle experiment	omit the minimax claim for richer-information settings

4 Algorithm

4.1 Centered leave-one-out base and raw randomized correction

Fix one frozen episode context and parameter, and suppress the episode index. A complete trajectory yields

$$\mathbf{X}_i = (V_i^+, V_i^-, G_i), \quad V_i^\pm = v_\pm(Y_i). \quad (27)$$

For $\sigma \in \{+, -\}$, let $\varsigma_+ = 1$, $\varsigma_- = -1$ and define

$$I_{i,\sigma}(y) = \mathbf{1}\{V_i^\sigma > y\}, \quad \hat{S}_{-i,n}^\sigma(y) = \frac{1}{n-1} \sum_{j \neq i} I_{j,\sigma}(y). \quad (28)$$

The centered leave-one-out statistic is

$$U_n^{\text{cen}} = \frac{1}{n} \sum_{i=1}^n G_i \sum_{\sigma \in \{+,-\}} \varsigma_\sigma \int_0^{B_v} (w_\sigma^\varepsilon)'(\hat{S}_{-i,n}^\sigma(y)) \{I_{i,\sigma}(y) - \hat{S}_{-i,n}^\sigma(y)\} dy. \quad (29)$$

The corresponding raw statistic is

$$U_n^{\text{raw}} = \frac{1}{n} \sum_{i=1}^n G_i \sum_{\sigma \in \{+,-\}} \varsigma_\sigma \int_0^{B_v} (w_\sigma^\varepsilon)'(\hat{S}_{-i,n}^\sigma(y)) I_{i,\sigma}(y) dy. \quad (30)$$

Choose $n \geq 2$, activation probability $\varrho \in (0, 1]$, and exponent $b \in (1, 2)$. Let

$$\nu_\ell = (1 - 2^{-b})2^{-b(\ell-1)}, \quad \ell = 1, 2, \dots \quad (31)$$

For $n_\ell = n2^\ell$, compute $U_{n_\ell}^{\text{raw}}$ on one batch of n_ℓ trajectories and the two half-sample statistics on its two disjoint halves. Define

$$D_{n,\ell}^{\text{raw}} = U_{n_\ell}^{\text{raw}} - \frac{1}{2} \left(U_{n_{\ell-1}}^{\text{raw},(1)} + U_{n_{\ell-1}}^{\text{raw},(2)} \right). \quad (32)$$

Let $A \sim \text{Bernoulli}(\varrho)$ and, conditional on $A = 1$, let $\ell^* \sim \nu$; both are independent of the base sample and the correction batch. One complete debiased output is

$$Z_n^{\text{db}} = U_n^{\text{cen}} + \frac{A}{\varrho \nu_{\ell^*}} D_{n,\ell^*}^{\text{raw}}, \quad (33)$$

with the second term defined as zero when $A = 0$.

Algorithm 1 One centered-LOO debiased CPT-gradient output

Require: $(z, r, \tilde{P}, \theta)$, $n \geq 2$, $\varrho \in (0, 1]$, $b \in (1, 2)$

- 1: $\mathcal{X}_0 \leftarrow \{\mathbf{X}_i\}_{i=1}^n$ i.i.d. from the frozen law at (z, r, θ) ; $U_0 \leftarrow U_n^{\text{cen}}(\mathcal{X}_0)$
 - 2: $A \sim \text{Bernoulli}(\varrho)$
 - 3: **if** $A = 0$ **then**
 - 4: **return** U_0
 - 5: **end if**
 - 6: $\ell \sim \nu$; $\mathcal{X}_\ell \leftarrow \{\mathbf{X}_i\}_{i=1}^{n2^\ell}$ i.i.d. from the same frozen law, independently of $\mathcal{X}_0$
 - 7: $D \leftarrow U_{n2^\ell}^{\text{raw}}(\mathcal{X}_\ell) - \frac{1}{2} \{U_{n2^{\ell-1}}^{\text{raw}}(\mathcal{X}_\ell^{(1)}) + U_{n2^{\ell-1}}^{\text{raw}}(\mathcal{X}_\ell^{(2)})\}$
 - 8: **return** $Z_n^{\text{db}} = U_0 + D/(\varrho \nu_\ell)$
-

The level distribution is untruncated: no deterministic cap is imposed on ℓ^* , and completed outputs are not discarded as a function of the realized level or magnitude.

4.2 Online update

In episode k , generate M_k conditionally independent copies of (33) under the frozen simulator and set

$$\hat{g}_k^{\text{db}} = \frac{1}{M_k} \sum_{m=1}^{M_k} Z_{k,m}^{\text{db}}. \quad (34)$$

The parameter update is

$$\theta_{k+1} = \Pi_\Theta \left\{ \theta_k + \gamma \mathbf{d}(\hat{g}_k^{\text{db}}) \right\}. \quad (35)$$

Algorithm 2 Dynamic-reference CPT policy gradient (DRCPT-PG)

Require: $\theta_1 \in \Theta$, (X_1, r_1, ω_0) , $h, \chi, \gamma, \vartheta$, (n, ϱ, b) , $\{M_k\}_{k=1}^K$

- 1: **for** $k = 1, \dots, K$ **do**
- 2: fix $(z_k, r_k, \tilde{P}_k)$ conditionally on $\mathcal{F}_k$
- 3: generate $Z_{k,1}^{\text{db}}, \dots, Z_{k,M_k}^{\text{db}}$ independently by Algorithm 1
- 4: $\hat{g}_k^{\text{db}} \leftarrow M_k^{-1} \sum_{m=1}^{M_k} Z_{k,m}^{\text{db}}$
- 5: **for** $u = t_k, \dots, t_k + h - 1$ **do**
- 6: $\tilde{\omega}_u \sim \pi_{\theta_k}(\cdot \mid s_u)$; $\omega_u \leftarrow (1 - \chi)\omega_{u-1} + \chi \mathbf{M}_u(\tilde{\omega}_u)$
- 7: $X_{u+1} \leftarrow X_u \{1 + \omega_u^\top R_{u+1} - c_{\text{tc}} \|\omega_u - \omega_{u-1}\|_1\}$; $r_{u+1} \leftarrow \Phi(r_u, X_{u+1})$
- 8: **end for**
- 9: $\theta_{k+1} \leftarrow \Pi_\Theta\{\theta_k + \gamma \mathbf{d}(\hat{g}_k^{\text{db}})\}$
- 10: **end for**
- 11: **return** $\{\theta_k, X_{t_k}, r_k, \omega_{t_k-1}\}_{k=1}^{K+1}$

5 Main Results

5.1 Reference dependence

Proposition 5.1 (Same terminal wealth, different CPT value). *Let $x_{\text{lo}} < r_0 < x_{\text{f}} < x_{\text{hi}}$ and consider*

$$\text{I} : r_0 \rightarrow x_{\text{hi}} \rightarrow x_{\text{f}}, \quad \text{II} : r_0 \rightarrow x_{\text{lo}} \rightarrow x_{\text{f}}.$$

Assume

$$x_{\text{f}} < (1 - \eta_+)r_0 + \eta_+x_{\text{hi}}, \quad x_{\text{f}} > (1 - \eta_-)r_0 + \eta_-x_{\text{lo}}.$$

Then

$$r_{\text{I}} - r_{\text{II}} = \eta_+(1 - \eta_-)x_{\text{hi}} - \eta_-(1 - \eta_+)x_{\text{lo}} + (\eta_- - \eta_+)x_{\text{f}}. \quad (36)$$

If the right-hand side is nonzero, the two paths have different regularized CPT values despite identical terminal wealth x_{f} .

Proposition 5.2 (Reference sensitivity). *Under Assumptions 3.1, 3.2, and 3.5, set*

$$\eta_{\min} = \min(\eta_+, \eta_-), \quad L_v = \alpha_+ \varepsilon_v^{\alpha_+ - 1} + \lambda \alpha_- \varepsilon_v^{\alpha_- - 1}. \quad (37)$$

For two episode-start references r, r' coupled through the same wealth path,

$$|r_{t_k+h}(r) - r_{t_k+h}(r')| \leq (1 - \eta_{\min})^h |r - r'|, \quad (38)$$

and, for fixed (P, z, θ) ,

$$|J(P, z, r; \theta) - J(P, z, r'; \theta)| \leq C_1 L_v (1 - \eta_{\min})^h |r - r'|. \quad (39)$$

Consequently, with

$$\begin{aligned} \Delta_k^{\text{nr}} &:= \sup_{\theta \in \Theta} |J(P_{k+1}, z_{k+1}, r_k; \theta) - J(P_k, z_k, r_k; \theta)|, \\ \sup_{\theta \in \Theta} |J_{k+1}(\theta, r_{k+1}) - J_k(\theta, r_k)| &\leq \Delta_k^{\text{nr}} + C_1 L_v (1 - \eta_{\min})^h |r_{k+1} - r_k|. \end{aligned} \quad (40)$$

5.2 Gradient identity and debiased estimation

Theorem 5.3 (Regularized CPT score-gradient identity). *Under Assumptions 3.1–3.2, fix one target and write $J = J_k$, $Y = Y_k$, $S_\theta^\pm(y) := S_{P_k, z_k, r_k, \theta}^\pm(y)$, and $G_\theta := G_{k, \theta}$. Define*

$$\begin{aligned} \Psi_\theta(\tau) = & \int_0^{B_v} (w_+^\varepsilon)'(S_\theta^+(y)) \{ \mathbf{1}[v_+(Y(\tau)) > y] - S_\theta^+(y) \} dy \\ & - \int_0^{B_v} (w_-^\varepsilon)'(S_\theta^-(y)) \{ \mathbf{1}[v_-(Y(\tau)) > y] - S_\theta^-(y) \} dy. \end{aligned} \quad (41)$$

Then

$$\nabla J(\theta) = \mathbb{E}\{\Psi_\theta(\tau)G_\theta(\tau)\}. \quad (42)$$

The same identity holds for the frozen simulator objective $\tilde{J}$.

Lemma 5.4 (Coupled full/half cancellation). *Under Assumptions 3.1–3.2, fix one target and parameter and write $g := \nabla J(\theta)$. There exist a measurable $\mathfrak{h}$ and constants $C_{\text{rem}}, C_U < \infty$ such that*

$$U_n^{\text{raw}} - g = \frac{1}{n} \sum_{i=1}^n \mathfrak{h}(\mathbf{X}_i) + \mathcal{B}_n, \quad \mathbb{E}\mathfrak{h}(\mathbf{X}_i) = 0, \quad \mathbb{E}\|\mathcal{B}_n\|_2^2 \leq \frac{C_{\text{rem}}}{n^2}, \quad (43)$$

which implies

$$\mathbb{E}\|D_{n, \ell}^{\text{raw}}\|_2^2 \leq \frac{9C_{\text{rem}}}{n^2} 2^{-2\ell}, \quad \ell \geq 1, \quad (44)$$

and

$$\text{tr Cov}(U_n^{\text{cen}}) \leq \frac{C_U}{n}. \quad (45)$$

Lemma 5.5 (Centering and finite-sample bias). *For $n \geq 2$,*

$$\mathbb{E}U_n^{\text{cen}} = \mathbb{E}U_n^{\text{raw}} =: \mu_n, \quad (46)$$

and there exists $C_{\text{bias}} < \infty$ such that

$$\|\mu_n - g\|_2 \leq \frac{C_{\text{bias}}}{n}, \quad g := \nabla J(\theta). \quad (47)$$

Theorem 5.6 (Exact unbiasedness, finite variance, and finite expected cost). *Let Z_n^{db} be defined by (33), with $n \geq 2$, $\varrho \in (0, 1]$, and $1 < b < 2$. Put*

$$\Xi_b := \sum_{\ell \geq 1} \frac{4^{-\ell}}{\nu_\ell} = \frac{1}{4(1 - 2^{-b})(1 - 2^{b-2})}, \quad \Upsilon_b := \sum_{\ell \geq 1} \nu_\ell 2^\ell = \frac{2(1 - 2^{-b})}{1 - 2^{1-b}}. \quad (48)$$

Then

$$\mathbb{E}Z_n^{\text{db}} = g, \quad (49)$$

$$\text{tr Cov}(Z_n^{\text{db}}) \leq C_{\text{var}}(n) := \frac{C_U}{n} + \frac{9C_{\text{rem}}\Xi_b}{\varrho n^2} < \infty, \quad (50)$$

and the expected number of complete trajectories is

$$C_{\text{cost}}(n, \varrho, b) = n(1 + \varrho\Upsilon_b) < \infty. \quad (51)$$

For M independent outputs,

$$\widehat{g}_{M, n}^{\text{db}} := \frac{1}{M} \sum_{m=1}^M Z_{n, m}^{\text{db}}, \quad \mathbb{E}\|\widehat{g}_{M, n}^{\text{db}} - g\|_2^2 \leq \frac{C_{\text{var}}(n)}{M}. \quad (52)$$

Theorem 5.7 (Fixed-target asymptotic normality). *Under Theorem 5.6, fix (n, ϱ, b) and let $\Sigma_n = \text{Cov}(Z_n^{\text{db}})$. Then*

$$\sqrt{M} (\hat{g}_{M,n}^{\text{db}} - g) \rightsquigarrow \mathcal{N}(0, \Sigma_n). \quad (53)$$

If Σ_n is nonsingular and $\hat{\Sigma}_{M,n}$ is the empirical covariance of the M outputs, then

$$\sqrt{M} \hat{\Sigma}_{M,n}^{-1/2} (\hat{g}_{M,n}^{\text{db}} - g) \rightsquigarrow \mathcal{N}(0, I_d). \quad (54)$$

5.3 Online residual, dynamic local regret, and stationary-set tracking

Lemma 5.8 (One-step normalized projected ascent). *Let J be L -smooth on the nonempty closed convex set Θ . Fix $x \in \Theta$, let $g = \nabla J(x)$ with $\|g\|_2 \leq B_g$, and let $\hat{g}$ be any finite vector. Set*

$$x^+ = \Pi_{\Theta}\{x + \gamma \mathbf{d}(\hat{g})\}, \quad Q(x) = \gamma^{-1}[\Pi_{\Theta}\{x + \gamma \mathbf{d}(g)\} - x].$$

If $\gamma L \leq \vartheta$ and $C_g := \max\{B_g, \vartheta\}$, then

$$J(x^+) - J(x) \geq \frac{\vartheta \gamma}{8} \|Q(x)\|_2^2 - \frac{5C_g \gamma}{4\vartheta^2} \|\hat{g} - g\|_2^2. \quad (55)$$

Theorem 5.9 (Average expected squared projected residual bound). *Suppose Assumptions 3.1–3.3 hold and $\gamma L \leq \vartheta$. Let $\hat{g}_k^{\text{db}}$ be defined by (34) and target $\tilde{g}_k(\theta_k)$. Then*

$$\mathbb{E} \|\hat{g}_k^{\text{db}} - g_k(\theta_k)\|_2^2 \leq \frac{C_{\text{var}}(n)}{M_k} + \mathbb{E} \beta_k^2, \quad (56)$$

and

$$\mathcal{E}_K \leq \frac{8\{2B_v + \mathbb{E} \mathcal{V}_K\}}{\vartheta \gamma K} + \frac{10C_g}{\vartheta^3 K} \sum_{k=1}^K \left(\frac{C_{\text{var}}(n)}{M_k} + \mathbb{E} \beta_k^2 \right), \quad (57)$$

Corollary 5.10 (Dynamic local regret). *For every $K \geq 1$, pathwise,*

$$\mathcal{R}_{\rho,w}(K) \leq \sum_{k=1}^K \zeta_k, \quad (58)$$

and therefore

$$\frac{1}{K} \mathbb{E} \mathcal{R}_{\rho,w}(K) \leq \mathcal{E}_K. \quad (59)$$

Corollary 5.11 (Moving stationary-set tracking). *If Assumption 3.4 also holds, then*

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E} \text{dist}^2(\theta_k, \mathbf{S}_k) \leq \kappa^2 \mathcal{E}_K. \quad (60)$$

Lemma 5.12 (Drift of stationary sets). *Under Assumption 3.4, define*

$$\delta_k^g := \sup_{\theta \in \Theta} \|g_{k+1}(\theta) - g_k(\theta)\|_2. \quad (61)$$

Then

$$d_{\text{H}}(\mathbf{S}_{k+1}, \mathbf{S}_k) \leq \frac{\kappa}{\vartheta} \delta_k^g. \quad (62)$$

Corollary 5.13 (Vanishing average expected squared projected residual and DLR). *Fix $w < \infty$ and $\rho \in (0, 1]$. If*

$$\mathbb{E}\mathcal{V}_K = o(K), \quad \frac{1}{K} \sum_{k=1}^K \left(\frac{1}{M_k} + \mathbb{E}\beta_k^2 \right) \rightarrow 0, \quad (63)$$

then

$$\mathcal{E}_K \rightarrow 0, \quad K^{-1} \mathbb{E}\mathcal{R}_{\rho, w}(K) \rightarrow 0. \quad (64)$$

Under Assumption 3.4, the left-hand side of (60) also converges to zero.

5.4 Fixed-dimension minimax information rate

Under Assumption 3.6, define

$$\mathfrak{R}_B := \inf_{\hat{g}: \sup_{P \in \mathfrak{C}} \mathbb{E}_P T_{\text{call}} \leq B} \sup_{P \in \mathfrak{C}} \mathbb{E}_P \|\hat{g} - g_P\|_2^2. \quad (65)$$

Theorem 5.14 (Minimax-optimal fixed-dimension trajectory-budget rate). *Under Assumption 3.6, there exist constants $c_{\text{lb}}, C_{\text{ub}}, B_0 \in (0, \infty)$, independent of B , such that for all $B \geq B_0$,*

$$\frac{c_{\text{lb}}}{B} \leq \mathfrak{R}_B \leq \frac{C_{\text{ub}}}{B}. \quad (66)$$

The upper bound is attained, up to constants, by averaging independent untruncated outputs Z_n^{db} under the same expected complete-trajectory budget. Hence, for fixed d and fixed regularization, the minimax gradient-estimation rate is $\Theta(B^{-1})$.

6 Experimental Design and Evidence

6.1 Experimental roadmap

The experiments follow the argument developed in the theory. E1 isolates the path-dependent reference mechanism and its decision consequence. E2 evaluates the frozen-target gradient estimator in the shared semi-synthetic setup. E3 releases the same setup as a continuous slow-variation process and tests online first-order tracking, while E4 maps the robustness of that tracking result over reference, horizon, and optimizer perturbations. E5 then moves to a time-ordered real A-share replay under common trading constraints. E6 holds real investor decisions, controls, chronological split, and predictive model fixed while changing only the reference representation. E7 replays the E5 recommendations through investor types calibrated from E6 and historical behavior, separating recommendation adoption from satisfaction after adoption. Figure 1 summarizes the theory-aligned core and these three external-validity layers.

Experimental evidence map

Mechanism / Statistical estimation / Online adaptation / Robustness / External validity

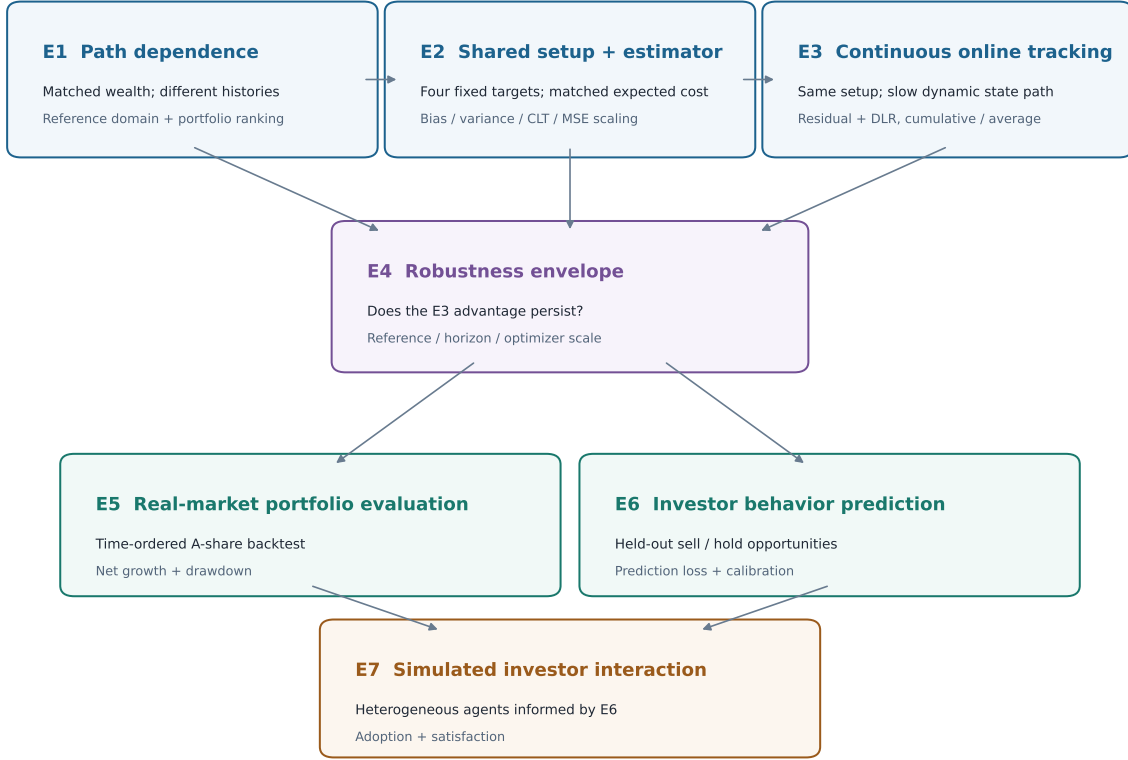


Figure 1: Experimental evidence map for the theory-aligned core. E1 establishes the decision consequence of path-dependent reference formation; E2 evaluates the frozen-target gradient estimator; E3 measures continuous first-order tracking; and E4 maps robustness. E5–E7 follow this core as external-validity layers for market, behavior, and interaction, while keeping their estimands separate from the E2–E4 statistical and tracking diagnostics.

6.2 E1: Path dependence and portfolio choice

Design. We construct two histories with the same initial wealth and the same endpoint,

$$(1, 1.10, 1.015) \quad \text{and} \quad (1, 0.90, 1.015),$$

and apply the reference recursion in (3) with $(\eta_+, \eta_-) = (0.20, 0.05)$. The histories are matched at the decision time: current wealth is $X_T = 1.015$, the pre-trade portfolio is 50% cash and 50% risky asset, and the next risky return is +4% with probability p and −4% otherwise. We compare two feasible target allocations, 20% and 80% risky. Under the action inertia in (7), these targets execute as 44% and 56% risky. Because the next-period distribution has two states, all CPT values are evaluated exactly rather than by Monte Carlo.

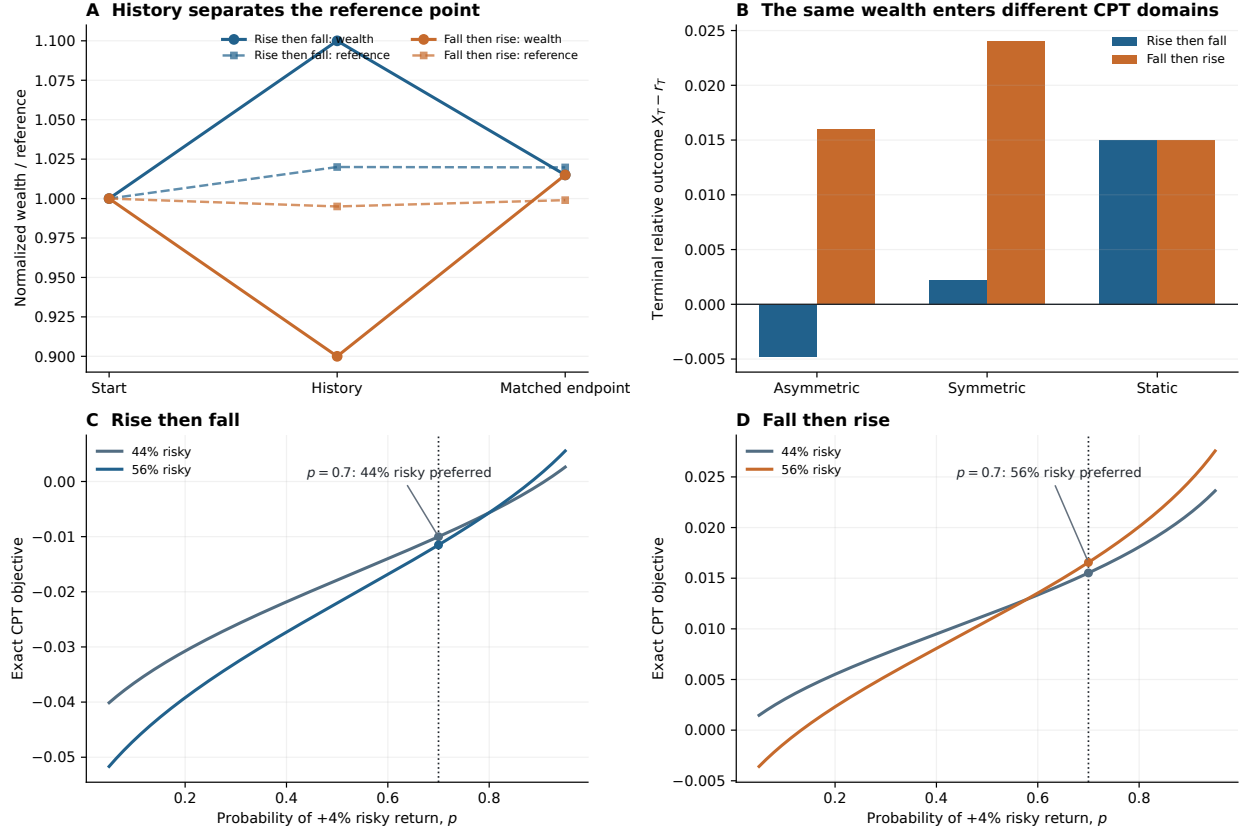


Figure 2: Path dependence changes both the perceived domain and the portfolio ranking. **A:** The two histories end at the same wealth but generate different asymmetric reference points. **B:** The asymmetric rule places the rise–fall history below its reference and the fall–rise history above its reference; the symmetric and static rules provide mechanism controls. **C–D:** Exact CPT values of the two feasible executed allocations as the probability p of a +4% risky return varies. At $p = 0.7$, the matched rise–fall account prefers 44% risky whereas the matched fall–rise account prefers 56% risky.

Table 2: E1 exact mechanism summary at $p = 0.7$.

History	X_T	r_T	$X_T - r_T$	Preferred risky share	CPT domain
Rise then fall	1.015	1.01975	−0.00475	44%	Loss
Fall then rise	1.015	0.99900	+0.01600	56%	Gain

Results. The key result is a rank reversal under a matched current state. The two accounts have identical terminal wealth, pre-trade holdings, and next-period opportunity set; only their realized histories differ. Asymmetric adaptation places the rise–fall account below its reference, $X_T - r_T = -0.00475$, but the fall–rise account above its reference, $X_T - r_T = 0.016$. The same $\{+4\%, -4\%$ risky-return distribution is therefore evaluated on opposite branches of the CPT value function. At $p = 0.7$, this changes the preferred executed risky share from 44% after the rise–fall path to 56% after the fall–rise path. The evidential point is not the absolute size of the four CPT values but the change in their ordering after current wealth, holdings, and opportunities have been

held fixed. Dynamic reference formation is therefore decision-relevant state information: a fixed current-state representation would collapse two histories that imply different feasible actions under the behavioral criterion. This is the decision-level consequence of the path dependence characterized in [Theorem 5.1](#).

6.3 E2: Finite-sample gradient estimation

Shared semi-synthetic setup. Beginning with E2, we use a common 30-stock-plus-cash semi-synthetic market built from a frozen empirical A-share factor-block library. Factor blocks are sampled from a time-invariant empirical distribution, while the return law follows a pre-specified path across episodes. For risky asset i in episode k ,

$$R_{i,k,u} = 0.08 \tanh\left(\frac{\mu_k + 2.5 b_k^\top f_{i,u} + \xi_{k,u}^{\text{com}} + \xi_{i,k,u}^{\text{id}}}{0.08}\right),$$

where the common and idiosyncratic shocks have standard deviations $0.006s_k$ and $0.009s_k$, respectively. Episodes 1–40 use the stable momentum law, episodes 41–60 use the reversal law, episodes 61–120 interpolate to the terminal low-volatility/value/quality law, and the exogenous return law is fixed thereafter. E2 standardizes the account context to

$$(X, r, \omega, \theta) = (1, 1, e_{\text{cash}}, 0)$$

and freezes four pre-specified targets at episodes 20, 50, 90, and 150, corresponding to the stable, post-change, mid-drift, and post-drift phases. E3 subsequently uses the same factor distribution and episode-specific return law but allows the account state to evolve continuously.

We compare the proposed centered leave-one-out estimator with independent raw randomized correction against the raw independent-split plug-in estimator used by the adapted CPT-PG v5 baseline. Both target the square-smoothed CPT objective in [\(9\)](#). Expected complete-trajectory cost is matched over

$$B \in \{32, 64, 128, 256, 512, 1024, 2048\},$$

and numerical gradient references are generated independently at substantially higher cost.

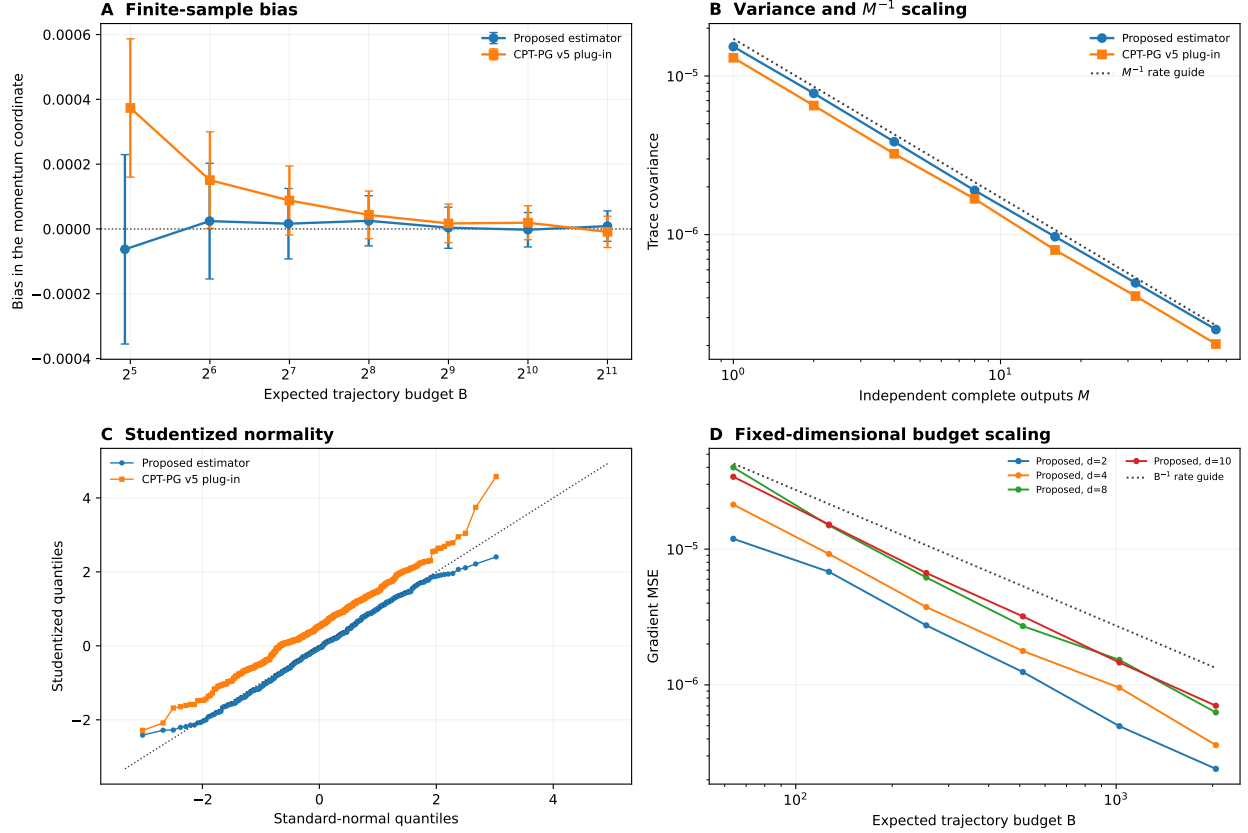


Figure 3: Finite-sample gradient estimation in the shared semi-synthetic setup. **A:** coordinate bias at the stable target. **B:** trace covariance of averages of M independent complete estimator outputs; the dotted reference has slope -1 . **C:** studentized normal-quantile diagnostic at $M = 64$. **D:** full-gradient MSE versus expected trajectory budget for $d \in \{2, 4, 8, 10\}$; the dotted reference has slope -1 . The corresponding unbiasedness, asymptotic normality, and fixed-dimensional minimax statements are given in [Theorems 5.6](#), [5.7](#) and [5.14](#).

Table 3: Full-gradient MSE at expected trajectory budget $B = 512$ across the four fixed targets. MSE entries are in units of 10^{-6} . The last column gives a paired 95% bootstrap interval for $\text{MSE}_{\text{prop}} - \text{MSE}_{\text{v5}}$ in the same units.

Target	Proposed	CPT-PG v5	Ratio	Difference 95% CI
Stable	3.123	2.938	1.063	$[-0.055, 0.434]$
Post-change	10.134	10.809	0.938	$[-1.495, 0.139]$
Mid-drift	10.647	11.468	0.928	$[-1.739, 0.102]$
Post-drift	10.589	12.257	0.864	$[-2.599, -0.721]$

Results. E2 separates three questions that an online gradient oracle must answer: centering, statistical efficiency, and inferential calibration. First, the proposed coordinate estimate stays centered near the numerical reference across the budget range, whereas the plug-in construction shows the low-budget displacement expected when nonlinear probability weighting is evaluated at an empirical distribution ([Figure 3A](#)). This is the finite-sample effect targeted by the leave-one-out

centering and independent randomized correction.

Second, removing that displacement does not destroy ordinary Monte Carlo efficiency. Averaging independent complete outputs gives fitted trace-covariance slopes of -0.989 for the proposed estimator and -0.999 for CPT-PG v5, and the proposed full-gradient MSE scales with budget at slopes between -1.17 and -1.12 across $d = 2, 4, 8, 10$ (Figure 3B,D). The correction therefore changes the estimator’s center while retaining the inverse-budget behavior predicted by Theorem 5.14. Third, the studentized statistic is close to its nominal Gaussian calibration: at $M = 64$ its mean is -0.049 , standard deviation 0.993 , and 95% coverage 96.25% , compared with mean 0.556 and coverage 91.75% for the v5 statistic (Figure 3C).

The four frozen targets show where this distinction matters for the full gradient. At the standard budget $B = 512$, the proposed estimator has the lower MSE point estimate in three targets, and after the completed drift its MSE is 13.6% lower with a paired interval entirely below zero (Table 3). Taken together, the evidence identifies the estimator’s role more precisely than a generic lower-variance claim: it removes a nonlinear finite-sample centering error while preserving the $1/B$ information rate, which is the property needed before the gradient is propagated through the online state dynamics.

6.4 E3: Continuous slow-variation online tracking

Design. E3 releases the market setup introduced in E2 as one continuous online process. No account state is reset. If a five-day episode starting from (X_k, r_k, ω_k) produces the terminal state $(\tilde{X}_{k+1}, \tilde{r}_{k+1}, \omega_{k+1}^{\text{end}})$, the next episode starts from the diminishing-gain transition

$$\begin{aligned}\log X_{k+1} &= (1 - \alpha_k) \log X_k + \alpha_k \log \tilde{X}_{k+1}, \\ r_{k+1} &= (1 - \alpha_k) r_k + \alpha_k \tilde{r}_{k+1}, \\ \omega_{k+1} &= (1 - \alpha_k) \omega_k + \alpha_k \omega_{k+1}^{\text{end}}, \quad \alpha_k = k^{-3/4}.\end{aligned}$$

The within-episode reference update remains the asymmetric mechanism in (3) with $(\eta_+, \eta_-) = (0.20, 0.05)$. Under the bounded analysis domain of Assumption 3.1 and the bounded semi-synthetic returns, the episode-start state increments are $O(\alpha_k)$, with

$$\sum_{k \leq K} \alpha_k = O(K^{1/4}) = o(K).$$

The factor-block distribution is time invariant, the exogenous return coefficients follow the E2 phase path through episode 120, and remain fixed for episodes 121–300. The finite-horizon sensitivity argument in Appendix D then converts these state increments into $\mathcal{V}_K \leq C \sum_{k < K} \alpha_k + C' = O(K^{1/4})$ on this common domain. This identifies the objective-level condition behind the slow-variation setting without removing the dynamic reference’s path dependence.

The comparison uses 20 independent test trajectories. The online learner and frozen control share the complete path through episode 40. Thereafter the online learner continues Algorithm 2, whereas the control freezes θ but continues to propagate wealth, reference, and holdings under the same slow transition. We fix $(\gamma, \vartheta) = (0.08, 0.05)$ before the E3 test runs and use the growing replication schedule

$$M_k = \max\{4, \lceil \sqrt{k} \rceil\},$$

for which $M_k \geq \sqrt{k}$ and $K^{-1} \sum_{k=1}^K M_k^{-1} \leq (2\sqrt{K} - 1)/K = O(K^{-1/2})$. DLR follows (19) with $w = 5$ and $\rho = 0.9$. Two independent evaluation streams estimate both the squared projected residual and the squared norm of the window-averaged projected residual by cross-stream products, preventing the ordinary self-square noise term from entering the evaluation.

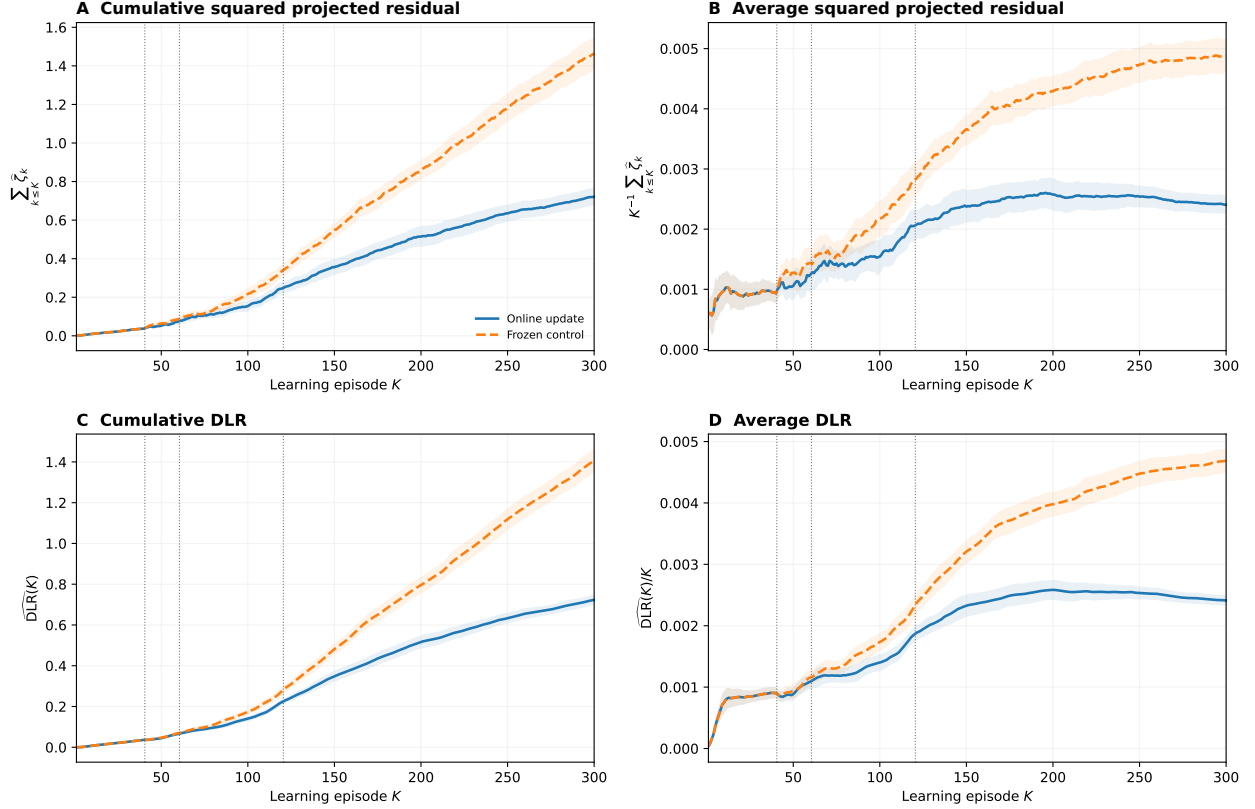


Figure 4: Continuous online tracking under slow variation. **A–B**: cumulative squared projected residual and its average. **C–D**: cumulative and average DLR. The two methods share the same trajectory through episode 40; thereafter the frozen control holds θ fixed while both methods continue to propagate their own wealth, reference, and holdings. Dotted lines mark the abrupt change, the start of gradual drift, and the point after which the exogenous return law is fixed. Curves are means over 20 independent trajectories and shaded bands are 1.96 standard errors.

Results. Continued adaptation changes the accumulation of first-order tracking error, not merely the terminal portfolio outcome. From episodes 41–300, the online learner accumulates 0.684 squared projected residual versus 1.427 for the frozen control, a reduction of about 52%; cumulative DLR is 0.687 versus 1.369, a reduction of about 50%. The corresponding seed-paired 95% bootstrap intervals for the differences are $[-0.834, -0.650]$ and $[-0.738, -0.622]$, and by episode 300 the average DLR is 2.411×10^{-3} online versus 4.685×10^{-3} frozen (Figure 4). Because the two policies generate different endogenous wealth, reference, and holding paths after episode 40, the frozen arm is deliberately a closed-loop control rather than a common-objective optimizer comparison. The relevant question is whether continued updating keeps each induced system closer to its own contemporaneous first-order condition, which is exactly what the projected residual and DLR measure.

The tail behavior strengthens this interpretation. After the exogenous return law becomes fixed, episodes 121–300 still contain slow endogenous state variation. Over this interval, cumulative DLR grows with empirical log–log slope 0.704 under continued updating but 0.968 under freezing; squared projected residual shows the same separation, 0.745 versus 0.959. Thus updating changes error accumulation from a nearly linear frozen trajectory toward the sublinear direction predicted by Theorems 5.10 and 5.13. The two independent evaluation streams also rule out the ordinary self-square noise term as an explanation for the separation. E3 therefore supports the mechanism

behind the online analysis: repeated first-order corrections absorb a slowly moving target rather than simply benefiting from a favorable realized account path. Under the optional error-bound condition in [Theorem 3.4](#), the same residual control additionally translates into average squared distance to the contemporaneous stationary set through [Theorem 5.11](#).

6.5 E4: Robustness of online tracking

Design. E4 tests whether the E3 closed-loop tracking advantage persists when the reference dynamics, decision horizon, and optimizer scale are perturbed. Because each perturbation changes the induced objective sequence, absolute DLR values are not compared across configurations. For configuration c and paired seed s , we form the matched-control ratio

$$\text{Ratio}_{\text{DLR}}(c, s) = \frac{\text{DLR}_{\text{online}}^{\text{post}}(c, s)}{\text{DLR}_{\text{frozen}}^{\text{post}}(c, s)},$$

where the post-change interval begins at the first regime change. Figures show the mean of the seed-level ratios with paired-seed bootstrap intervals, rather than the ratio of two across-seed mean DLR values. A ratio below one means that continued updating accumulates less DLR than freezing θ on that matched trajectory.

Panel A evaluates the full 9×9 reference grid

$$\eta_+, \eta_- \in \{0, 0.05, 0.10, \dots, 0.40\}.$$

Panel B varies the episode horizon over

$$h \in \{1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50\},$$

with the 200-, 300-, 600-, and 1500-day boundaries assigned to the nearest complete episode where needed; the state-gain clock remains tied to elapsed market time. Panel C evaluates 48 optimizer configurations retained by a pre-specified diagnostic screen from

$$\gamma \in \{0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.08, 0.10, 0.12\},$$

$$\vartheta \in \{0.005, 0.01, 0.015, 0.02, 0.03, 0.04, 0.05, 0.06\}, \quad \gamma L_{\text{diag}} \leq \vartheta, \quad L_{\text{diag}} = 0.31.$$

Here L_{diag} defines the displayed grid and is distinct from the uniform smoothness constant L in [Theorem 5.9](#). The E3 defaults $(\eta_+, \eta_-) = (0.20, 0.05)$, $h = 5$, and $(\gamma, \vartheta) = (0.08, 0.05)$ are marked but are not re-selected in E4. Each configuration uses four independent paired trajectories and two independent evaluation streams.

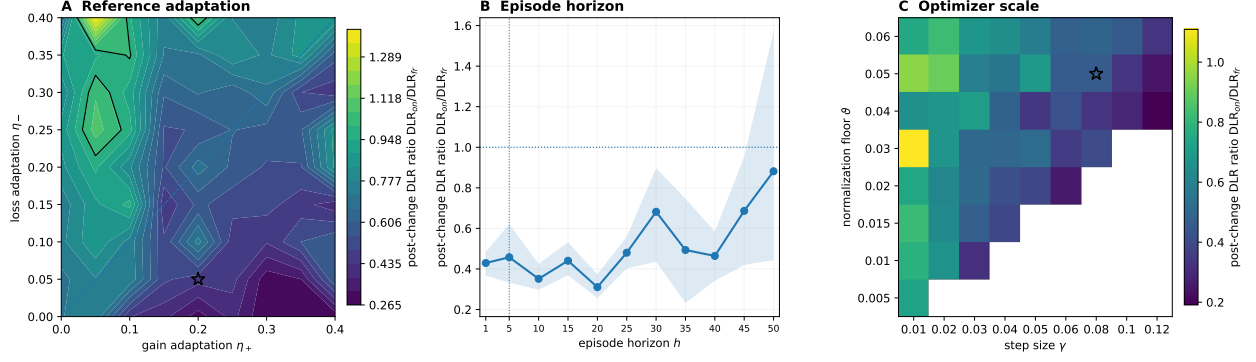


Figure 5: Robustness of the online-tracking advantage measured by the configuration-matched post-change DLR ratio $\text{Ratio}_{\text{DLR}}$. **A**: reference-adaptation surface over the simulated 9×9 grid; the black contour marks a ratio of one, the diagonal marks $\eta_+ = \eta_-$, and the star marks the default asymmetric rule. **B**: horizon sweep with paired-seed 95% bootstrap intervals; the horizontal line marks a ratio of one and the vertical line marks $h = 5$. **C**: diagnostically screened (γ, ϑ) grid; blank cells lie outside $\gamma L_{\text{diag}} \leq \vartheta$, and the star marks the E3 default.

Results. E4 asks whether the E3 separation is a property of the mechanism or an accident of one tuning choice. The reference sweep gives the clearest answer. Continued updating has lower post-change DLR in 76 of 81 reference configurations, and all 36 gain-dominant asymmetric settings with $\eta_+ > \eta_-$ lie below the matched-control boundary; 33 of those have paired 95% intervals entirely below one. The default rule has $\text{Ratio}_{\text{DLR}} = 0.458$ with interval $[0.332, 0.621]$. The surface also identifies a coherent boundary of the phenomenon: strongly loss-dominant adaptation can reverse the comparison, as at $(\eta_+, \eta_-) = (0.05, 0.40)$ where the ratio is 1.403 with interval $[1.224, 1.552]$ (Figure 5A). The robustness region therefore follows the qualitative reference dynamics rather than looking like isolated favorable cells.

The other two sweeps test whether the effect is tied to discretization or optimizer tuning. Mean ratios are below one at all 11 horizons, with 10 intervals entirely below one; only the $h = 50$ interval crosses one, so the E3 result is not specific to the five-day episode length (Figure 5B). Across the 48 screened (γ, ϑ) pairs, 47 mean ratios are below one and 45 intervals are entirely below one; the lone point estimate above one has interval $[0.876, 1.342]$ and is unresolved from the matched control. Taken together, the sweeps map a broad region in which continued adaptation improves first-order tracking. This is stronger evidence than a single default run because it shows that the E3 separation survives changes to the behavioral state dynamics, the temporal aggregation scale, and the optimizer itself.

6.6 E5: Real-market portfolio evaluation

Design. E5 transfers the portfolio learner from the controlled E2–E4 market to a chronological A-share replay. The universe is fixed before the test period at 300 HS300 constituents. Historical inputs begin on January 3, 2023, and the out-of-sample execution window runs from January 2, 2025 through May 29, 2026. Every factor is point-in-time: momentum, reversal, volatility, valuation, liquidity, size, and previous holdings use only information available before the execution date, and accounting-quality variables enter only after their announcement dates. The learner and the realized account use the same stochastic Dirichlet actor, and transaction cost $0.001 \|\omega_t - \omega_{t-1}\|_1$ is deducted from wealth.

We compare DRCPT-PG with symmetric-reference CPT-PG, static-reference CPT-PG, expected-wealth PG, exponential-utility PG, and a passive equal-weight anchor. The five learned policies use the same five-day horizon, $\gamma = 0.08$, expected trajectory budget 512, and ten pre-specified policy seeds; DRCPT-PG keeps $(\eta_+, \eta_-) = (0.20, 0.05)$. The primary market diagnostics are wealth, annualized return and volatility, Sharpe ratio, maximum drawdown, turnover, and cumulative transaction cost. DLR is not used to rank the real-market methods because the objective of E5 is economic external validity rather than theorem verification.

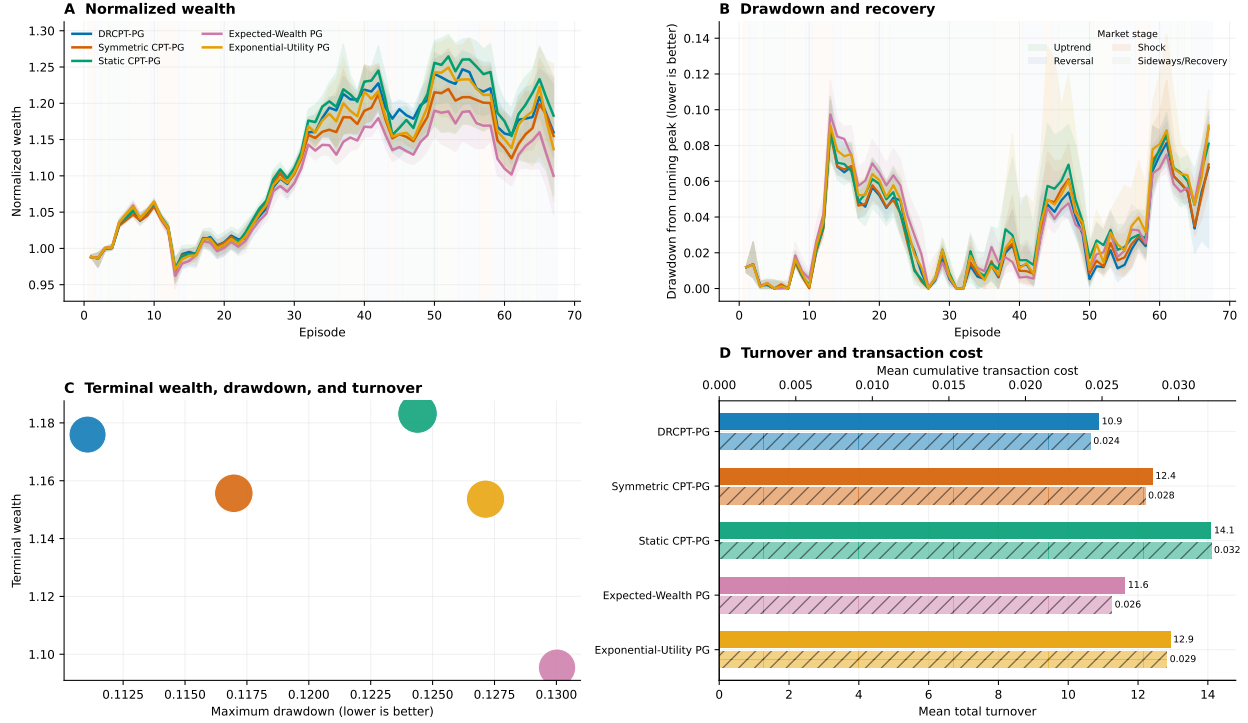


Figure 6: E5 chronological A-share evaluation. **A**: median normalized wealth across the ten policy seeds, with seed dispersion bands. **B**: drawdown and recovery over the same replay, with market-stage annotations used only for visualization. **C**: terminal wealth versus maximum drawdown; bubble size represents turnover. **D**: mean turnover and cumulative transaction cost. All learned methods share the same fixed asset universe, point-in-time information set, transaction-cost rule, and stochastic execution protocol.

Table 4: E5 real-market summary. Entries are means across the ten policy seeds; the equal-weight row is the deterministic passive anchor.

Method	Ann. return (%)	Sharpe	Max DD (%)	Turnover	Cost (%)
DRCPT-PG	13.03	0.856	11.67	10.87	2.42
Symmetric CPT-PG	13.21	0.796	12.48	12.41	2.78
Static CPT-PG	14.69	0.789	14.36	14.09	3.21
Expected-Wealth PG	10.51	0.606	14.23	11.63	2.56
Exponential-Utility PG	14.79	0.773	13.95	12.92	2.92
Equal Weight	16.64	1.068	11.59	1.00	0.20

Results. Across the learned policies, DRCPT-PG occupies the strongest average risk-adjusted and trading-efficiency profile in the replay. Its mean annualized return is 13.03% with annualized volatility 15.62%, giving the highest learned-policy Sharpe ratio, 0.856. It also has the lowest learned-policy maximum drawdown, 11.67%, and the lowest total turnover, 10.87, with cumulative transaction cost 2.42% (Table 4). Static CPT-PG and exponential-utility PG realize higher mean gross returns but do so in a region with larger drawdown, turnover, and cost, which is visible in the return–risk–turnover panel of Figure 6. The equal-weight portfolio is retained as a passive market anchor rather than a policy-gradient ablation. Thus the real-market separation of DRCPT-PG is a risk-adjusted and trading-intensity result: the endogenous reference mechanism changes how the learned policy uses risk and turnover under chronological A-share data, not merely its terminal wealth.

6.7 E6: Investor behavior prediction

Design. E6 asks whether the reference state used by the portfolio model carries incremental information about real investor decisions. The sample contains 306,444 held-stock rebalance opportunities, with 71,854 observations in the final chronological test split. The outcome is whether the investor decreases the stock’s target weight at the next rebalance. Four logistic specifications use exactly the same observations, outcome, controls, and train–validation–test dates; they differ only in the reference representation: no reference, static purchase cost, symmetric adaptation, or asymmetric adaptation with $(\eta_+, \eta_-) = (0.40, 0.10)$. Common controls include holding age, previous weight, price, prior investor reduction behavior, and lagged market, portfolio, and social aggregates. Uncertainty is assessed by 200 paired investor-cluster bootstrap draws.

The behavioral statistic is recomputed under each reference representation rather than fixing gain and loss states to the static purchase cost. For the same realized decisions, we therefore report the reduction rate when the position is above its own reference, the reduction rate when it is below its own reference, and their difference. This separates two questions: whether a reference representation improves held-out prediction and whether the gain/loss partition that it induces aligns with the observed disposition asymmetry.

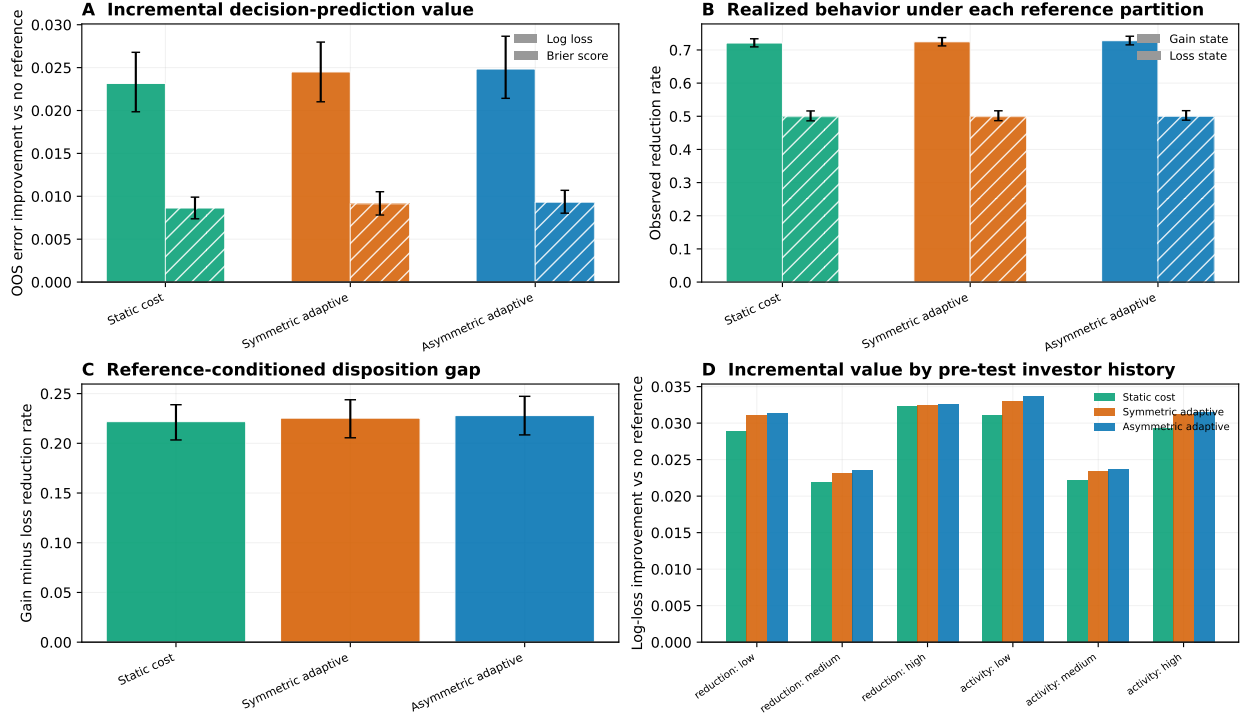


Figure 7: E6 held-out behavioral alignment. **A**: paired-bootstrap improvements in out-of-sample log loss and Brier score relative to the no-reference model. **B**: observed reduction rates in gain and loss states defined separately by each reference. **C**: reference-conditioned gain-minus-loss reduction gap. **D**: log-loss improvement over the no-reference model across pre-test investor-history strata. Error bars in A–C are paired investor-cluster 95% bootstrap intervals.

Results. The asymmetric dynamic reference is the best of the four matched predictive specifications on every reported test metric. On the 71,854 held-out decisions it achieves log loss 0.57937, Brier score 0.18696, AUROC 0.78124, and AUPRC 0.80909, compared with 0.60420, 0.19623, 0.76002, and 0.79060 without a reference. In the paired bootstrap, the asymmetric-minus-none difference is -0.02488 for log loss with 95% interval $[-0.02866, -0.02142]$, -0.00931 for Brier score with interval $[-0.01070, -0.00803]$, $+0.02127$ for AUROC with interval $[0.01862, 0.02394]$, and $+0.01852$ for AUPRC with interval $[0.01588, 0.02126]$.

The comparison with symmetric adaptation tests the incremental predictive value of gain–loss asymmetry beyond merely allowing the reference to move. The paired asymmetric-minus-symmetric log-loss difference is -0.000391 with 95% interval $[-0.000695, -0.000014]$, while the AUROC difference is $+0.000412$ with interval $[0.000164, 0.000664]$. Its reference-conditioned gain and loss reduction rates are 0.7278 and 0.5004, giving a disposition gap of 0.2274; the paired-bootstrap gap is 0.2279 with interval $[0.2085, 0.2473]$. The asymmetric gap exceeds the symmetric one by 0.00255 with interval $[0.00107, 0.00404]$. The same specification also gives the largest log-loss improvement in all six strata formed from pre-test reduction history and trading activity (Figure 7D). E6 therefore links the E1 reference mechanism to actual held-out decisions: asymmetric reference adaptation contributes stable incremental predictive information, while the estimand remains behavioral alignment rather than causal identification.

6.8 E7: Simulated heterogeneous-investor interaction

Design. E7 adds an interaction layer without retraining the portfolio policies. We calibrate four investor types from 3,481 real users using training-period E6 gain/loss reduction statistics together with historical activity, diversification, sentiment, and network summaries. The resulting population shares are 49.4% moderate-frequency focused investors, 28.7% disposition-effect investors, 19.4% active diversified investors, and 2.5% low-frequency diversified investors. Sixteen calibrated users are sampled per type, producing 64 simulated agents. Type construction uses training-period behavioral summaries only; the global E6 predictive coefficients are not clustering features.

For every method, the experiment exactly replays the full cash-plus-300-stock recommendations saved by E5. At the beginning of each five-day episode, adoption is sampled from the current recommendation, calibrated investor history, compatibility, and lagged satisfaction before any return from the episode is observed. If the recommendation is adopted, the agent executes that method for five days; otherwise it keeps the episode-start portfolio. Both paths apply their own trading costs and reference updates. Conditional satisfaction is the five-day normalized prospect-value improvement over the no-rebalance counterfactual, and interaction value multiplies that satisfaction by adoption so that rejected recommendations contribute zero. Seed-level market paths and adoption draws are paired across methods, and population summaries are weighted by the empirical type shares.

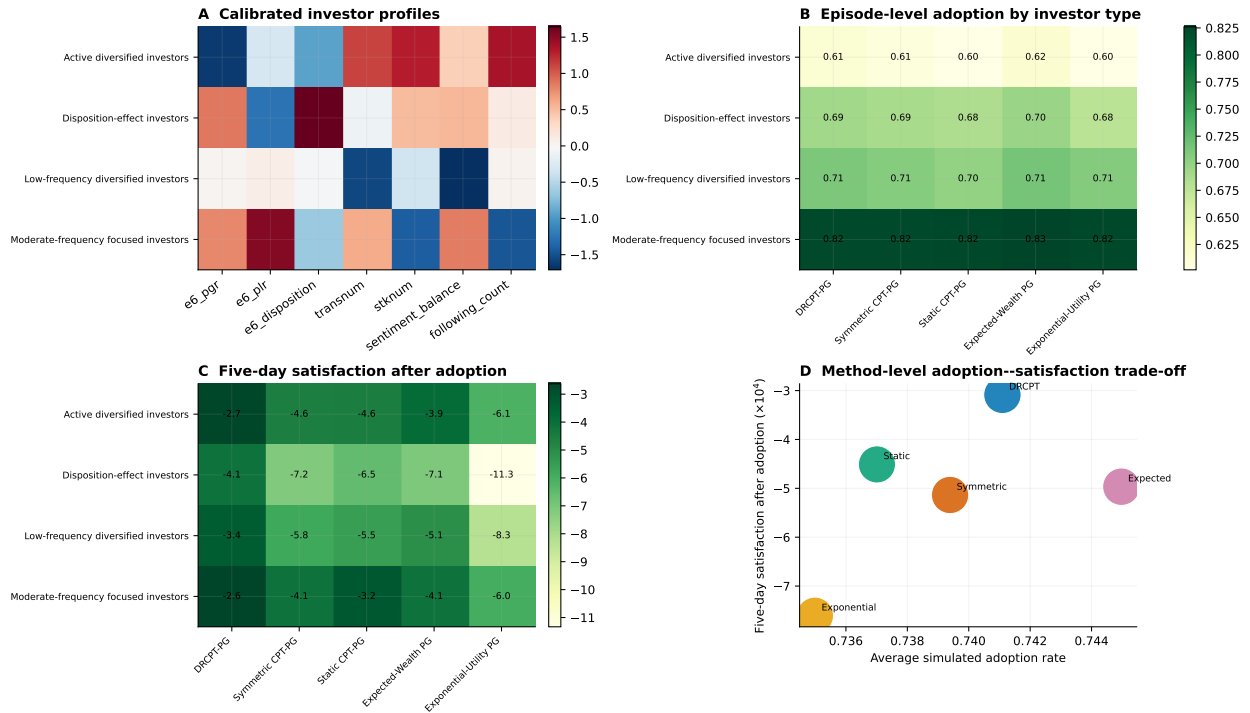


Figure 8: E7 calibrated heterogeneous-agent interaction. **A**: standardized profiles of the four investor types. **B**: simulated episode-level adoption by type and method. **C**: five-day conditional satisfaction after adoption. **D**: method-level adoption-satisfaction plane; bubble size encodes mean recommendation compatibility. Adoption is decided before episode returns are revealed, and satisfaction is evaluated against a no-rebalance counterfactual with its own reference path.

Results. The interaction layer separates recommendation uptake from value after uptake. DRCPT-PG has a population-weighted adoption rate of 0.741 and mean compatibility of 0.747. Its conditional

five-day satisfaction is -3.087×10^{-4} and its adoption-weighted interaction value is -2.265×10^{-4} . Among the learned advisers, both are the largest values: the interaction values are -3.237×10^{-4} for static CPT-PG, -3.739×10^{-4} for symmetric CPT-PG, -3.662×10^{-4} for expected-wealth PG, and -5.496×10^{-4} for exponential-utility PG. Expected-wealth PG has a slightly higher adoption rate (0.745) but lower post-adoption satisfaction, illustrating why adoption and satisfaction are kept as separate axes in [Figure 8D](#).

The learned-adviser ordering is also stable across heterogeneous types. DRCPT-PG has the highest adoption-weighted interaction value among the learned methods for active diversified, disposition-effect, low-frequency diversified, and moderate-frequency focused investors, despite their different activity, diversification, and reference profiles. It also uses lower mean turnover (9.29) and cumulative transaction cost (0.0208) than the other learned advisers in the interaction replay. The passive equal-weight anchor is shown separately in the same figure; the E7 claim concerns relative behavior within the adaptive learned-adviser family. Because adoption and satisfaction are calibrated counterfactual quantities, E7 is interpreted as a heterogeneity stress test of the recommendation layer, complementing the observed-decision evidence in E6.

7 Discussion

The results support one central conclusion: once the reference point is treated as an endogenous state, CPT portfolio learning becomes a coupled behavioral, statistical, and online-optimization problem. E1 shows why the state is consequential—matched accounts can rank the same feasible portfolios differently solely because their wealth histories imply different references. E2–E4 then show that the corresponding first-order target can be estimated with controlled statistical error and tracked under a moving objective. E5–E7 carry the same mechanism into chronological market data, held-out investor decisions, and heterogeneous interaction without replacing the theorem-aligned diagnostics by downstream performance measures.

Relative to fixed-reference CPT policy-gradient optimization, the main technical change is therefore not simply an additional state variable. Endogenous reference adaptation makes the objective path dependent and moving, while nonlinear probability weighting makes its frozen-target gradient statistically nontrivial. The centered leave-one-out base and independent randomized correction address the latter by producing exact unbiasedness under the complete-trajectory oracle, and the projected-residual analysis addresses the former by replacing single-point convergence with first-order tracking. The estimator diagnostics, slow-variation experiment, and dense robustness sweep jointly support this separation: centering, trajectory-budget scaling, online residual accumulation, and sensitivity are measured on the objects that appear in the theory.

The external-validity layers reveal a complementary pattern. In the real A-share replay, DRCPT-PG’s separation is strongest in risk-adjusted performance, drawdown, and turnover among learned policies rather than in gross return alone. In the investor data, asymmetric adaptation improves held-out prediction over no reference, static reference, and symmetric adaptation under the same observations and controls, and it induces the largest reference-conditioned disposition gap. The heterogeneous-agent replay then shows why a recommendation layer needs two behavioral axes: adoption can be high even when post-adoption value differs, and DRCPT-PG attains the largest adoption-weighted satisfaction among learned advisers across all four calibrated types. These results connect dynamic reference formation to both trading intensity and decision behavior while preserving distinct economic and behavioral estimands.

The evidence map also defines the scope of the claims. The estimator and tracking theorems concern the regularized frozen-target gradient and a slowly varying objective sequence under the

stated assumptions. E5 evaluates chronological market transfer with quote-relative returns and common trading costs; E6 is a held-out predictive comparison rather than a causal identification design; and E7 is a calibrated counterfactual interaction assay rather than observed user adoption or satisfaction. Keeping these layers separate prevents portfolio returns, predictive fit, and simulated interaction value from becoming interchangeable proxies for first-order optimization quality. A natural next step is a prospective human-in-the-loop evaluation in which reference adaptation, recommendation uptake, realized portfolio outcomes, and user-reported utility are collected under a common pre-registered protocol.

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A Notation and Analytic Preliminaries

All fixed-target statistical statements condition on the episode-start information, the queried parameter, and the frozen simulator before new Monte Carlo trajectories are drawn. Conditional independence is always with respect to this law. Limits in M fix the target and estimator design (n, ϱ, b) ; limits in K require the constants in the stated assumptions to be uniform over episodes; the minimax limit fixes d and all model-class constants while $B \rightarrow \infty$.

A.1 Main notation

Table 5: Main notation along the state, objective, estimation, and online-evaluation chain. Estimator internals and proof-local constants are defined where they are used.

Symbol	Type	Meaning
k, u, h	indices / integer	episode, calendar time, and finite episode horizon
$s_u, \omega_u, \tilde{\omega}_u$	state / portfolios	state $(m_u, X_u, r_u, \omega_{u-1})$ with market state m_u ; executed portfolio and latent proposal
$z_k, \mathcal{F}_k, r_k$	context / information	frozen non-reference start state, episode-start information, and reference r_{t_k}
$\theta, \Theta, \pi_\theta$	vector / set / policy	parameter, compact convex parameter set, and Dirichlet actor
$P_k, \tilde{P}_k$	episode models	true and frozen-simulator models inducing policy-dependent trajectory laws
$Y_k, v_\pm, w_\pm^\varepsilon$	outcome / functions	terminal relative outcome and regularized CPT value and weight maps
$J_k, \tilde{J}_k$	scalar functions	true and frozen-simulator CPT objectives
$g_k, \tilde{g}_k, G_{k,\theta}$	$\mathbb{R}^d$	true gradient, simulator gradient, and complete-trajectory score
$\hat{g}_k^{\text{db}}, M_k$	vector / integer	debaised update estimate averaged over M_k independent complete outputs
$Q_k; \gamma, \vartheta$	vector / scalars	projected residual; step size and normalization floor
S_k	subset of Θ	constrained first-order stationary set
ζ_k	$\mathbb{R}_+$	squared projected residual $\ Q_k(\theta_k)\ _2^2$
$\mathcal{E}_K$	$\mathbb{R}_+$	average expected squared projected residual
$\mathcal{R}_{\rho,w}(K); \rho, w$	scalar / parameters	dynamic local regret (DLR); discount and window length
$\beta_k, \mathcal{V}_K$	$\mathbb{R}_+$	simulator-gradient error bound and cumulative true-objective variation
T_{call}, B	integer / scalar	oracle trajectory count and expected trajectory budget

A.2 Dirichlet score moments

Lemma A.1 (Uniform moments of the augmented trajectory score). *Under Assumption 3.1, there exist $0 < \underline{a} \leq \bar{a} < \infty$ such that*

$$\underline{a} \leq a_{i,u}(\theta) \leq \bar{a} \quad (67)$$

uniformly in i, u, θ . For every fixed $p < \infty$,

$$\sup_{k,\theta} \mathbb{E} \|G_{k,\theta}\|_2^p < \infty. \quad (68)$$

The trajectory log-likelihood Hessian $\mathcal{H}_{k,\theta}$ also satisfies

$$\sup_{k,\theta} \mathbb{E} \|\mathcal{H}_{k,\theta}\|_{\text{op}} < \infty. \quad (69)$$

Moreover, for every bounded measurable trajectory functional F , differentiation under the conditional trajectory law is valid up to order two:

$$\nabla_{\theta} \mathbb{E}_{\theta} F = \mathbb{E}_{\theta}(F G_{k,\theta}), \quad \nabla_{\theta}^2 \mathbb{E}_{\theta} F = \mathbb{E}_{\theta}\{F(G_{k,\theta} G_{k,\theta}^{\top} + \mathcal{H}_{k,\theta})\}. \quad (70)$$

Proof. Compactness of Θ and boundedness of $q_{i,u}$ imply (67). Every Dirichlet coordinate has a Beta marginal with parameters in a compact subset of $(0, \infty)^2$. Hence, for every finite p ,

$$\sup_{a,b} \int_0^1 |\log x|^p x^{a-1} (1-x)^{b-1} dx < \infty,$$

where the supremum is over the induced compact parameter set. The digamma and polygamma terms are bounded on the same positive compact interval. Equation (8), bounded features, and finite (h, N) therefore give (68); differentiating the score once more gives (69). The market kernel and $\mathbf{M}_u$ contribute no likelihood derivative by Assumption 3.1.

To justify differentiation under the trajectory integral, enlarge Θ slightly while keeping it compactly contained in the policy domain. On this neighborhood the Dirichlet concentrations remain in $[a_-, a_+]$ for some $a_- > 0$. For one latent action $\tilde{\omega}$, the Dirichlet density and its first two θ -derivatives are bounded, up to constants independent of θ , by

$$\prod_{i=0}^N \tilde{\omega}_i^{a_- - 1} \left(1 + \sum_{i=0}^N |\log \tilde{\omega}_i|^2 \right).$$

This envelope is integrable on the simplex for $a_- > 0$. Taking the finite product over the h latent actions gives an integrable trajectory envelope; the parameter-independent market kernels integrate to one conditionally at each step. Hence bounded F times either derivative is dominated by a common integrable function. Dominated convergence yields (70). $\square$

A.3 Smoothness of the CPT objective

Lemma A.2 (Uniform gradient and smoothness bounds). *Under Assumptions 3.1–3.2, there exist $B_g, L < \infty$ such that*

$$\sup_{k,\theta \in \Theta} \|g_k(\theta)\|_2 \leq B_g, \quad \|g_k(\theta) - g_k(\theta')\|_2 \leq L \|\theta - \theta'\|_2. \quad (71)$$

With

$$C_{G,2} := \sup_{k,\theta} \mathbb{E} \|G_{k,\theta}\|_2^2, \quad C_H := \sup_{k,\theta} \mathbb{E} \|\mathcal{H}_{k,\theta}\|_{\text{op}},$$

one admissible choice is

$$B_g = 2B_v C_1 \sqrt{C_{G,2}}, \quad L = 2B_v \{C_2 C_{G,2} + C_1 (C_{G,2} + C_H)\}. \quad (72)$$

Moreover $|J_k(\theta, r_k)| \leq B_v$.

Proof. For one side σ and threshold y , the likelihood-score identity and Lemma A.1 give

$$\nabla S_\theta^\sigma(y) = \mathbb{E}\{I_\sigma(y)G_\theta\}, \quad \nabla^2 S_\theta^\sigma(y) = \mathbb{E}\{I_\sigma(y)(G_\theta G_\theta^\top + \mathcal{H}_\theta)\}.$$

The bounded indicator and the moment bounds provide an integrable dominating function, so the derivative–expectation interchange is justified by dominated convergence. Consequently

$$\|\nabla S_\theta^\sigma(y)\|_2 \leq \sqrt{C_{G,2}}, \quad \|\nabla^2 S_\theta^\sigma(y)\|_{\text{op}} \leq C_{G,2} + C_H.$$

Differentiating the CPT integral twice and using Assumption 3.2 gives

$$\nabla^2 \int_0^{B_v} w_\sigma^\varepsilon(S_\theta^\sigma(y)) dy = \int_0^{B_v} \{(w_\sigma^\varepsilon)''(S) \nabla S \nabla S^\top + (w_\sigma^\varepsilon)'(S) \nabla^2 S\} dy.$$

The integrand is uniformly dominated by the constants above, giving (72). The first-derivative bound is analogous. Finally $0 \leq w_\sigma^\varepsilon \leq 1$ and the two integrals have common length B_v , so $|J_k| \leq B_v$. $\square$

A.4 Projected residual and constrained stationarity

Lemma A.3 (Projected residual characterizes first-order stationarity). *For closed convex Θ , $\gamma, \vartheta > 0$, and $g \in \mathbb{R}^d$,*

$$\Pi_\Theta\{\theta + \gamma d(g)\} = \theta \iff \langle g \rangle u - \theta \leq 0 \quad \forall u \in \Theta. \quad (73)$$

Hence the two definitions in (16) coincide.

Proof. The projection variational inequality gives $x = \Pi_\Theta(y)$ iff $\langle y - x \rangle u - x \leq 0$ for every $u \in \Theta$. Substitute $y = \theta + \gamma d(g)$ and use $d(g) = g / \max\{\|g\|_2, \vartheta\}$, a positive scalar multiple of g . $\square$

B Proofs for the Reference Mechanism and CPT Gradient

B.1 Same terminal wealth and path-dependent reference

Proof. It is enough to compute the two terminal references and then use strict monotonicity of the deterministic CPT value.

For path I, the first step is a gain, so its interim reference is

$$r_I^{(1)} = (1 - \eta_+)r_0 + \eta_+x_{\text{hi}}.$$

The first displayed condition in the proposition makes the second step a loss relative to this interim reference. Hence

$$r_I = (1 - \eta_-)\{(1 - \eta_+)r_0 + \eta_+x_{\text{hi}}\} + \eta_-x_f.$$

For path II, the first step is a loss and

$$r_{II}^{(1)} = (1 - \eta_-)r_0 + \eta_-x_{\text{lo}}.$$

The second displayed condition makes the move to x_f a gain, so

$$r_{II} = (1 - \eta_+)\{(1 - \eta_-)r_0 + \eta_-x_{\text{lo}}\} + \eta_+x_f.$$

Subtracting cancels the coefficient of r_0 and gives (36).

It remains to translate a nonzero reference gap into a CPT-value gap. A deterministic relative outcome y_0 has survival function equal to one below its corresponding value magnitude and zero above it. Since $w_\pm^\varepsilon(0) = 0$ and $w_\pm^\varepsilon(1) = 1$, its CPT value is $v_+(y) - v_-(y)$. The shifted-power gain term is strictly increasing on $y > 0$, and $-v_-(y)$ is strictly increasing on $y < 0$; the two pieces meet continuously at zero. Thus the signed deterministic CPT value is strictly increasing in y . If $r_I \neq r_{II}$, then $x_f - r_I \neq x_f - r_{II}$, so the two CPT values differ. $\square$

B.2 Reference contraction and CPT sensitivity

Proof. To prove (39), it is enough to establish three bounds: contraction of the reference recursion, Lipschitz continuity of the value transformation, and an L^1 survival-function coupling inequality.

Fix a wealth value x . The map $r \mapsto \Phi(r, x)$ is continuous, with slope $1 - \eta_+$ when $r \leq x$ and slope $1 - \eta_-$ when $r > x$. Splitting an interval at x when necessary therefore gives

$$|\Phi(r, x) - \Phi(r', x)| \leq (1 - \eta_{\min})|r - r'|.$$

Under Assumption 3.5, the two compared trajectories have the same wealth path. Iterating the preceding inequality for h steps proves (38). Since their terminal wealth is common,

$$|Y(r) - Y(r')| \leq (1 - \eta_{\min})^h |r - r'|. \quad (74)$$

For $x \geq 0$,

$$\frac{d}{dx} u_{\alpha, \varepsilon_v}(x) = \alpha x(x^2 + \varepsilon_v^2)^{\alpha/2-1} \leq \alpha \varepsilon_v^{\alpha-1},$$

where the bound uses $x \leq (x^2 + \varepsilon_v^2)^{1/2}$ and $\alpha \leq 1$. Hence

$$|v_+(y) - v_+(y')| + |v_-(y) - v_-(y')| \leq L_v |y - y'|.$$

Combining this with (74) controls the sum of the two terminal value-magnitude differences.

Finally, if $V, V' \in [0, B_v]$ are coupled nonnegative random variables, Tonelli's theorem gives

$$\begin{aligned} \int_0^{B_v} |P(V > y) - P(V' > y)| dy &\leq \mathbb{E} \int_0^{B_v} |\mathbf{1}\{V > y\} - \mathbf{1}\{V' > y\}| dy \\ &= \mathbb{E}|V - V'|. \end{aligned}$$

Since $|w_\sigma^\varepsilon(p) - w_\sigma^\varepsilon(q)| \leq C_1 |p - q|$, applying this inequality to the gain and loss integrals and using the preceding value bound proves (39). The decomposition (40) follows by adding and subtracting $J(P_{k+1}, z_{k+1}, r_k; \theta)$ and applying the triangle inequality uniformly in θ . $\square$

B.3 Score representation of the CPT gradient

Proof. We prove the gain part; the loss part follows by the same calculation and is subtracted at the end. To differentiate

$$J^+(\theta) = \int_0^{B_v} w_+^\varepsilon(S_\theta^+(y)) dy,$$

it is enough to identify $\nabla S_\theta^+(y)$ and justify moving the derivative through the y -integral.

For fixed y , the likelihood-ratio identity in Theorem A.1 applied to the bounded indicator gives

$$\nabla S_\theta^+(y) = \mathbb{E}\{\mathbf{1}[v_+(Y) > y] G_\theta\}. \quad (75)$$

By Cauchy-Schwarz and Assumption 3.1, the norm of the right-hand side is at most $\sqrt{C_{G,2}}$ uniformly in y and θ . Since $(w_+^\varepsilon)'$ is bounded by C_1 and the integration interval is finite, this constant is an integrable dominating function. The dominated convergence theorem therefore gives

$$\begin{aligned} \nabla J^+(\theta) &= \int_0^{B_v} (w_+^\varepsilon)'(S_\theta^+(y)) \mathbb{E}\{\mathbf{1}[v_+(Y) > y] G_\theta\} dy \\ &= \mathbb{E} \left[G_\theta \int_0^{B_v} (w_+^\varepsilon)'(S_\theta^+(y)) \mathbf{1}[v_+(Y) > y] dy \right], \end{aligned}$$

where Fubini's theorem is applicable because the integrand is dominated by $C_1 B_v \|G_\theta\|$ and $\mathbb{E}\|G_\theta\| < \infty$.

The score has mean zero. Therefore subtracting the deterministic quantity $S_\theta^+(y)$ inside the integral does not change the expectation. This yields the gain component of (41). Repeating the argument for the loss side and subtracting proves (42). The frozen-simulator identity follows under the same conditional argument with its trajectory law in place of the true law. $\square$

C Proofs for the Debiased Estimator

Throughout this appendix, condition on one frozen target and write $g = \nabla J(\theta)$. The complete trajectory summaries $\mathbf{X}_1, \mathbf{X}_2, \dots$ are independent and identically distributed under this conditional law. For a realization $x = (V^+, V^-, G_x)$, write $I_{x,\sigma}(y) = \mathbf{1}\{V^\sigma > y\}$.

C.1 Second-order expansion and full/half coupling

Proof of Theorem 5.4. To prove (43), it is enough to isolate the first-order Hoeffding projection and show that both the degenerate two-sample term and the Taylor remainder are $O_{L^2}(n^{-1})$. For a side σ and threshold y , write

$$S_\sigma(y) = P(V^\sigma > y), \quad e_{-i,\sigma}(y) = \widehat{S}_{-i,n}^\sigma(y) - S_\sigma(y).$$

Taylor's theorem applied to $(w_\sigma^\varepsilon)'$ gives

$$(w_\sigma^\varepsilon)'(S_\sigma + e) = (w_\sigma^\varepsilon)'(S_\sigma) + (w_\sigma^\varepsilon)''(S_\sigma)e + \text{Rem}_\sigma(e), \quad |\text{Rem}_\sigma(e)| \leq \frac{C_3}{2}e^2. \quad (76)$$

Substituting (76) into (30) splits the statistic into a first-order trajectory term, an ordered two-sample term, and a Taylor remainder.

Define

$$\begin{aligned} h_0(x) &= G_x \sum_\sigma \varsigma_\sigma \int_0^{B_v} (w_\sigma^\varepsilon)'(S_\sigma(y)) I_{x,\sigma}(y) dy, \\ \mathcal{K}(x, x') &= G_x \sum_\sigma \varsigma_\sigma \int_0^{B_v} (w_\sigma^\varepsilon)''(S_\sigma(y)) I_{x,\sigma}(y) \{I_{x',\sigma}(y) - S_\sigma(y)\} dy. \end{aligned}$$

By (42) and $\mathbb{E}G = 0$, $\mathbb{E}h_0(\mathbf{X}) = g$. Moreover $\mathbb{E}\{\mathcal{K}(\mathbf{X}, \mathbf{X}') \mid \mathbf{X}\} = 0$ for independent $\mathbf{X}, \mathbf{X}'$, because $\mathbb{E}\{I_{\mathbf{X}',\sigma}(y) - S_\sigma(y)\} = 0$. Let

$$h_1(x') = \mathbb{E}_{\mathbf{X}}\{\mathcal{K}(\mathbf{X}, x')\}.$$

Then $\mathbb{E}h_1(\mathbf{X}') = 0$. Put $\mathcal{K}^\circ(x, x') = \mathcal{K}(x, x') - h_1(x')$. Both first-order projections of $\mathcal{K}^\circ$ vanish:

$$\mathbb{E}\{\mathcal{K}^\circ(\mathbf{X}, \mathbf{X}') \mid \mathbf{X}\} = 0, \quad \mathbb{E}\{\mathcal{K}^\circ(\mathbf{X}, \mathbf{X}') \mid \mathbf{X}'\} = 0.$$

The ordered two-sample term therefore has the exact decomposition

$$\frac{1}{n(n-1)} \sum_{i \neq j} \mathcal{K}(\mathbf{X}_i, \mathbf{X}_j) = \frac{1}{n} \sum_{j=1}^n h_1(\mathbf{X}_j) + \mathcal{D}_n, \quad (77)$$

where

$$\mathcal{D}_n = \frac{1}{n(n-1)} \sum_{i \neq j} \mathcal{K}^\circ(\mathbf{X}_i, \mathbf{X}_j).$$

The kernel is square-integrable because the indicators are bounded, the integration interval is finite, $(w_\sigma^\varepsilon)''$ is bounded, and $\mathbb{E}\|G\|_2^2 < \infty$. Degeneracy implies that covariances between two ordered kernel terms vanish unless their ordered index pairs coincide or are reversals of one another. Counting these $O(n^2)$ contributing pairs against the normalization $n^2(n-1)^2$ gives

$$\mathbb{E}\|\mathcal{D}_n\|_2^2 \leq \frac{C_D}{n^2} \quad (78)$$

for a finite constant C_D .

We next bound the Taylor remainder. For each i , $e_{-i,\sigma}$ is an empirical mean based on $n-1$ Bernoulli variables independent of $\mathbf{X}_i$. Uniformly in y ,

$$\mathbb{E}|e_{-i,\sigma}(y)|^4 \leq \frac{C_e}{(n-1)^2} \quad (79)$$

for a universal finite C_e . By (76), Cauchy–Schwarz in the y -integral, independence of G_i from the leave-one-out empirical survival, and $\mathbb{E}\|G_i\|_2^2 < \infty$, the i th Taylor contribution has second moment at most C_T/n^2 . Jensen’s inequality for the average over i then gives a second-moment bound of the same order for the averaged Taylor remainder T_n :

$$\mathbb{E}\|T_n\|_2^2 \leq \frac{C'_T}{n^2}. \quad (80)$$

Combining the first-order term, (77), and the Taylor remainder yields

$$U_n^{\text{raw}} - g = \frac{1}{n} \sum_{i=1}^n \mathfrak{h}(\mathbf{X}_i) + \mathcal{B}_n, \quad \mathfrak{h} = h_0 - g + h_1, \quad \mathcal{B}_n = \mathcal{D}_n + T_n.$$

The vector function $\mathfrak{h}$ has mean zero, and (78)–(80) imply $\mathbb{E}\|\mathcal{B}_n\|_2^2 \leq C_{\text{rem}}/n^2$ after increasing the finite constant C_{rem} . This proves (43).

Now take $n_\ell = n2^\ell$ and evaluate the full-sample statistic and its two half-sample statistics on the same n_ℓ trajectories. Applying (43) to the full sample and to each half gives

$$U_{n_\ell}^{\text{raw}} = g + \frac{1}{n_\ell} \sum_{i=1}^{n_\ell} \mathfrak{h}(\mathbf{X}_i) + \mathcal{B}_{n_\ell},$$

and

$$\frac{1}{2}\{U_{n_\ell/2}^{\text{raw},(1)} + U_{n_\ell/2}^{\text{raw},(2)}\} = g + \frac{1}{n_\ell} \sum_{i=1}^{n_\ell} \mathfrak{h}(\mathbf{X}_i) + \frac{1}{2}\{\mathcal{B}_{n_\ell/2}^{(1)} + \mathcal{B}_{n_\ell/2}^{(2)}\}.$$

The influence terms cancel exactly. Hence

$$D_{n,\ell}^{\text{raw}} = \mathcal{B}_{n_\ell} - \frac{1}{2}\{\mathcal{B}_{n_\ell/2}^{(1)} + \mathcal{B}_{n_\ell/2}^{(2)}\}.$$

Minkowski’s inequality in L^2 gives

$$(\mathbb{E}\|D_{n,\ell}^{\text{raw}}\|_2^2)^{1/2} \leq \frac{\sqrt{C_{\text{rem}}}}{n_\ell} + \frac{1}{2} \frac{2\sqrt{C_{\text{rem}}}}{n_\ell} + \frac{1}{2} \frac{2\sqrt{C_{\text{rem}}}}{n_\ell} = \frac{3\sqrt{C_{\text{rem}}}}{n_\ell}.$$

Squaring and substituting $n_\ell = n2^\ell$ proves (44).

It remains to prove (45). View U_n^{cen} as a symmetric function of $\mathbf{X}_1, \dots, \mathbf{X}_n$, and let $U_n^{\text{cen},(i)}$ be the statistic after replacing $\mathbf{X}_i$ by an independent copy $\mathbf{X}'_i$. If G'_i denotes the score component of $\mathbf{X}'_i$,

write $L_f = C_1 + C_2$. The coefficient in one centered LOO summand is bounded in absolute value by $2B_v C_1$, so the contribution from the replaced summand has norm at most

$$\frac{2B_v C_1}{n} (\|G_i\|_2 + \|G'_i\|_2).$$

For $j \neq i$, replacing $\mathbf{X}_i$ changes each leave-one-out survival entering the j th summand by at most $(n-1)^{-1}$ pointwise. Since

$$\left| \frac{\partial}{\partial p} \{ (w_\sigma^\varepsilon)'(p)(I-p) \} \right| \leq L_f,$$

the scalar coefficient multiplying G_j changes by at most $2B_v L_f / (n-1)$. Hence, for a deterministic constant C_* depending only on B_v, C_1, C_2 ,

$$\|U_n^{\text{cen}} - U_n^{\text{cen},(i)}\|_2 \leq \frac{C_*}{n} (\|G_i\|_2 + \|G'_i\|_2) + \frac{C_*}{n(n-1)} \sum_{j \neq i} \|G_j\|_2.$$

Using $(\sum_{j \neq i} x_j)^2 \leq (n-1) \sum_{j \neq i} x_j^2$ and the uniform second-moment bound for the score, there is a finite constant C_E such that

$$\mathbb{E} \|U_n^{\text{cen}} - U_n^{\text{cen},(i)}\|_2^2 \leq \frac{C_E}{n^2}$$

uniformly in i . The vector Efron–Stein inequality yields

$$\text{tr Cov}(U_n^{\text{cen}}) = \mathbb{E} \|U_n^{\text{cen}} - \mathbb{E} U_n^{\text{cen}}\|_2^2 \leq \frac{1}{2} \sum_{i=1}^n \mathbb{E} \|U_n^{\text{cen}} - U_n^{\text{cen},(i)}\|_2^2 \leq \frac{C_U}{n}$$

with $C_U = C_E/2$. This completes the proof. $\square$

C.2 Centering and finite-sample bias

Proof of Theorem 5.5. To compare the two LOO statistics, subtract (29) from (30). The difference is

$$U_n^{\text{raw}} - U_n^{\text{cen}} = \frac{1}{n} \sum_{i=1}^n G_i H_{-i}^{\text{ctr}}, \quad (81)$$

where

$$H_{-i}^{\text{ctr}} = \sum_{\sigma \in \{+, -\}} \varsigma_\sigma \int_0^{B_v} (w_\sigma^\varepsilon)'(\hat{S}_{-i,n}^\sigma(y)) \hat{S}_{-i,n}^\sigma(y) dy.$$

The coefficient H_{-i}^{ctr} depends only on trajectories with indices different from i . It is therefore independent of G_i . The score identity gives $\mathbb{E} G_i = 0$, while Assumptions 3.1–3.2 make $G_i H_{-i}^{\text{ctr}}$ integrable. Hence every summand in (81) has mean zero and $\mathbb{E} U_n^{\text{cen}} = \mathbb{E} U_n^{\text{raw}} = \mu_n$.

It remains to control $\mu_n - g$. This follows from the expansion in Theorem 5.4. Taking expectations in (43) and using $\mathbb{E} h(\mathbf{X}) = 0$ yields

$$\|\mu_n - g\|_2 = \|\mathbb{E} \mathcal{B}_n\|_2 \leq (\mathbb{E} \|\mathcal{B}_n\|_2^2)^{1/2} \leq \frac{\sqrt{C_{\text{rem}}}}{n}.$$

Thus (47) holds with $C_{\text{bias}} = \sqrt{C_{\text{rem}}}$. $\square$

C.3 Unbiasedness, variance, cost, and the CLT

Proof of Theorem 5.6. We first prove exact unbiasedness. By Theorem 5.5, the centered base has mean μ_n . For the raw correction,

$$\mathbb{E}D_{n,\ell}^{\text{raw}} = \mu_{n2^\ell} - \mu_{n2^{\ell-1}}. \quad (82)$$

The level bound (44) implies

$$\sum_{\ell \geq 1} \mathbb{E} \|D_{n,\ell}^{\text{raw}}\|_2 \leq \frac{3\sqrt{C_{\text{rem}}}}{n} \sum_{\ell \geq 1} 2^{-\ell} < \infty,$$

so the expectation can be interchanged with the infinite randomized series. Since (47) gives $\mu_{n2^{\ell_{\max}}} \rightarrow g$,

$$\begin{aligned} \mathbb{E}Z_n^{\text{db}} &= \mu_n + \sum_{\ell \geq 1} \mathbb{E}D_{n,\ell}^{\text{raw}} \\ &= \mu_n + \lim_{\ell_{\max} \rightarrow \infty} (\mu_{n2^{\ell_{\max}}} - \mu_n) = g. \end{aligned}$$

This proves (49).

For the second moment of the correction, independence of the activation and level variables gives

$$\begin{aligned} \mathbb{E} \left\| \frac{A}{\varrho \nu_{\ell^*}} D_{n,\ell^*}^{\text{raw}} \right\|_2^2 &= \frac{1}{\varrho} \sum_{\ell \geq 1} \frac{\mathbb{E} \|D_{n,\ell}^{\text{raw}}\|_2^2}{\nu_\ell} \\ &\leq \frac{9C_{\text{rem}}}{\varrho n^2} \sum_{\ell \geq 1} \frac{4^{-\ell}}{\nu_\ell} = \frac{9C_{\text{rem}}\Xi_b}{\varrho n^2}. \end{aligned}$$

The series Ξ_b is finite because $b < 2$. The base sample and correction sample are independent, so their covariance is zero. Combining the correction bound with (45) gives (50).

One base output costs exactly n complete trajectories. Conditional on activation and level ℓ , the correction uses one fresh batch of $n2^\ell$ trajectories; the two halves reuse this same batch. Hence

$$\mathbb{E} \text{cost}(Z_n^{\text{db}}) = n + \varrho n \sum_{\ell \geq 1} \nu_\ell 2^\ell = n(1 + \varrho \Upsilon_b),$$

where $\Upsilon_b < \infty$ because $b > 1$. Direct summation of the geometric series gives the closed forms in (48).

Finally, if $Z_{n,1}^{\text{db}}, \dots, Z_{n,M}^{\text{db}}$ are independent copies, exact unbiasedness and independence imply

$$\mathbb{E} \|\hat{g}_{M,n}^{\text{db}} - g\|_2^2 = \frac{1}{M} \text{tr Cov}(Z_n^{\text{db}}) \leq \frac{C_{\text{var}}(n)}{M}.$$

This is (52). □

Proof of Theorem 5.7. For fixed target and fixed (n, ϱ, b) , Theorem 5.6 gives iid vectors $Z_{n,r}^{\text{db}}$ with mean g and finite covariance Σ_n . For any fixed $t \in \mathbb{R}^d$, the scalar variables $t^\top (Z_{n,r}^{\text{db}} - g)$ are iid with finite variance. The classical scalar central limit theorem therefore gives

$$\sqrt{M} t^\top (\hat{g}_{M,n}^{\text{db}} - g) \rightsquigarrow \mathcal{N}(0, t^\top \Sigma_n t).$$

The Cramér–Wold device proves (53).

Because $\mathbb{E}\|Z_n^{\text{db}}\|_2^2 < \infty$, every entry of $Z_n^{\text{db}}(Z_n^{\text{db}})^\top$ is integrable. The law of large numbers therefore implies $\widehat{\Sigma}_{M,n} \rightarrow \Sigma_n$ in probability entrywise, hence in operator norm in fixed dimension. If Σ_n is nonsingular, operator-norm consistency implies $\mathbb{P}\{\lambda_{\min}(\widehat{\Sigma}_{M,n}) > \lambda_{\min}(\Sigma_n)/2\} \rightarrow 1$. On this event the inverse square-root map is well defined and continuous in a neighborhood of Σ_n . Slutsky’s theorem applied to (53) then proves (54). $\square$

D Proofs for the Online Residual and Tracking Results

D.1 Normalized projected ascent

Proof of Theorem 5.8. The target is a lower bound in terms of the population projected residual $Q(x)$. We first obtain a bound in terms of the actual displacement produced by $\widehat{g}$, then compare that displacement with the population displacement.

Let

$$\Delta x = x^+ - x, \quad c_g = \max\{\|g\|_2, \vartheta\}, \quad \mathbf{e} = \mathbf{d}(\widehat{g}) - \mathbf{d}(g).$$

The projection variational inequality with the feasible comparison point x gives

$$\langle \mathbf{d}(\widehat{g}), \Delta x \rangle \geq \frac{\|\Delta x\|_2^2}{\gamma}. \quad (83)$$

Since $g = c_g \mathbf{d}(g)$, the L -smooth lower quadratic bound and (83) imply

$$\begin{aligned} J(x + \Delta x) - J(x) &\geq \langle g, \Delta x \rangle - \frac{L}{2} \|\Delta x\|_2^2 \\ &= c_g \langle \mathbf{d}(\widehat{g}) - \mathbf{e}, \Delta x \rangle - \frac{L}{2} \|\Delta x\|_2^2 \\ &\geq \left(\frac{c_g}{\gamma} - \frac{L}{2} \right) \|\Delta x\|_2^2 - c_g \|\mathbf{e}\|_2 \|\Delta x\|_2 \\ &\geq \left(\frac{3c_g}{4\gamma} - \frac{L}{2} \right) \|\Delta x\|_2^2 - c_g \gamma \|\mathbf{e}\|_2^2 \\ &\geq \frac{c_g}{4\gamma} \|\Delta x\|_2^2 - c_g \gamma \|\mathbf{e}\|_2^2. \end{aligned} \quad (84)$$

The penultimate line uses $uv \leq u^2/(4\gamma) + \gamma v^2$, and the last line uses $\gamma L \leq \vartheta \leq c_g$.

Now let

$$t = \Pi_{\Theta}\{x + \gamma \mathbf{d}(g)\} - x = \gamma Q(x).$$

Nonexpansiveness of Euclidean projection gives $\|\Delta x - t\|_2 \leq \gamma \|\mathbf{e}\|_2$. From $\|t\|_2^2 \leq 2\|\Delta x\|_2^2 + 2\|\Delta x - t\|_2^2$,

$$\|\Delta x\|_2^2 \geq \frac{1}{2} \|t\|_2^2 - \gamma^2 \|\mathbf{e}\|_2^2.$$

Substituting into (84) yields

$$J(x^+) - J(x) \geq \frac{c_g \gamma}{8} \|Q(x)\|_2^2 - \frac{5c_g \gamma}{4} \|\mathbf{e}\|_2^2.$$

Finally, $\mathbf{d}(g) = \Pi_{\{u: \|u\|_2 \leq 1\}}(g/\vartheta)$, so nonexpansiveness of the unit-ball projection implies

$$\|\mathbf{e}\|_2 \leq \vartheta^{-1} \|\widehat{g} - g\|_2.$$

Using $c_g \geq \vartheta$ in the positive term and $c_g \leq C_g = \max(B_g, \vartheta)$ in the negative term proves (55). $\square$

D.2 Average expected squared projected residual

Proof of Theorem 5.9. We first connect the final estimator to the true gradient. Conditional on $\mathcal{F}_k$, Theorem 5.6 gives

$$\mathbb{E}(\hat{g}_k^{\text{db}} - \tilde{g}_k(\theta_k) \mid \mathcal{F}_k) = 0, \quad \mathbb{E}(\|\hat{g}_k^{\text{db}} - \tilde{g}_k(\theta_k)\|_2^2 \mid \mathcal{F}_k) \leq \frac{C_{\text{var}}(n)}{M_k}.$$

The cross term with the deterministic conditional vector $\tilde{g}_k(\theta_k) - g_k(\theta_k)$ therefore vanishes. Hence

$$\begin{aligned} \mathbb{E}\|\hat{g}_k^{\text{db}} - g_k(\theta_k)\|_2^2 &= \mathbb{E}\|\hat{g}_k^{\text{db}} - \tilde{g}_k(\theta_k)\|_2^2 + \mathbb{E}\|\tilde{g}_k(\theta_k) - g_k(\theta_k)\|_2^2 \\ &\leq \frac{C_{\text{var}}(n)}{M_k} + \mathbb{E}\beta_k^2, \end{aligned}$$

which is (56).

We next apply Theorem 5.8 to the true episode objective J_k at $x = \theta_k$ and $\hat{g} = \hat{g}_k^{\text{db}}$. Let

$$\mathbf{j}_k = J_k(\theta_{k+1}) - J_k(\theta_k).$$

After taking expectations,

$$\frac{\vartheta\gamma}{8}\mathbb{E}\zeta_k \leq \mathbb{E}\mathbf{j}_k + \frac{5C_g\gamma}{4\vartheta^2}\mathbb{E}\|\hat{g}_k^{\text{db}} - g_k(\theta_k)\|_2^2. \quad (85)$$

The objective increments telescope across moving objectives as the pathwise identity

$$\begin{aligned} \sum_{k=1}^K \mathbf{j}_k &= J_K(\theta_{K+1}) - J_1(\theta_1) \\ &\quad + \sum_{k=1}^{K-1} \{J_k(\theta_{k+1}) - J_{k+1}(\theta_{k+1})\}. \end{aligned} \quad (86)$$

Since $|J_k| \leq B_v$ and the second line is bounded above by $\mathcal{V}_K$,

$$\sum_{k=1}^K \mathbf{j}_k \leq 2B_v + \mathcal{V}_K. \quad (87)$$

Summing (85), applying (56) and (87), and dividing by $\vartheta\gamma K/8$ gives exactly (57). $\square$

D.3 Discounted dynamic local regret

Proof of Theorem 5.10. Write $W = W_{\rho,w}$. With the convention $Q_t(\theta_t) = 0$ for $t \leq 0$, the coefficients ρ^j/W , $j = 0, \dots, w-1$, are nonnegative and sum to one. Convexity of $q \mapsto \|q\|_2^2$ therefore gives

$$\|\bar{Q}_k\|_2^2 \leq \frac{1}{W} \sum_{j=0}^{w-1} \rho^j \|Q_{k-j}(\theta_{k-j})\|_2^2.$$

Summing over $k \leq K$ and reindexing,

$$\mathcal{R}_{\rho,w}(K) \leq \sum_{t=1}^K \zeta_t \left\{ \frac{1}{W} \sum_{j=0}^{\min\{w-1, K-t\}} \rho^j \right\} \leq \sum_{t=1}^K \zeta_t,$$

which proves (58). Taking expectations and dividing by K gives (59). $\square$

D.4 Stationary-set consequences

Proof of Theorem 5.11. Assumption 3.4 gives pointwise

$$\text{dist}^2(\theta_k, \mathcal{S}_k) \leq \kappa^2 \|Q_k(\theta_k)\|_2^2 = \kappa^2 \zeta_k.$$

Taking expectations and averaging over k proves (60). Substitution of (57) yields the stated finite-horizon control. $\square$

Proof of Theorem 5.12. The normalized direction map is ϑ^{-1} -Lipschitz because it is the Euclidean projection of g/ϑ onto the closed unit ball. Using nonexpansiveness of Π_Θ in (15),

$$\begin{aligned} \sup_{\theta \in \Theta} \|Q_{k+1}(\theta) - Q_k(\theta)\|_2 &\leq \sup_{\theta \in \Theta} \|\mathbf{d}(g_{k+1}(\theta)) - \mathbf{d}(g_k(\theta))\|_2 \\ &\leq \frac{\delta_k^g}{\vartheta}. \end{aligned}$$

If $\theta \in \mathcal{S}_k$, then $Q_k(\theta) = 0$, so Assumption 3.4 applied to episode $k+1$ gives

$$\text{dist}(\theta, \mathcal{S}_{k+1}) \leq \kappa \|Q_{k+1}(\theta)\|_2 \leq \frac{\kappa \delta_k^g}{\vartheta}.$$

Taking the supremum over $\theta \in \mathcal{S}_k$ gives one directed Hausdorff bound. Interchanging k and $k+1$ gives the reverse bound because the same uniform gradient-difference bound is symmetric. Taking the maximum proves (62). $\square$

Proof of Theorem 5.13. In (57), the endpoint term is $O(K^{-1})$. The assumption $\mathbb{E}\mathcal{V}_K = o(K)$ makes the variation term vanish after division by K . Since $C_{\text{var}}(n)$ is fixed,

$$\frac{1}{K} \sum_{k=1}^K \left(\frac{C_{\text{var}}(n)}{M_k} + \mathbb{E}\beta_k^2 \right) \rightarrow 0$$

by the assumed Cesàro condition. Therefore $\mathcal{E}_K \rightarrow 0$. Equation (59) gives the DLR conclusion, and (60) gives the stationary-set conclusion when Assumption 3.4 is imposed. $\square$

Objective variation in the semi-synthetic environment. Write $z_k = (\log X_k, r_k, \omega_k)$ and $\chi_k = (\mu_k, b_k, s_k)$ for the episode-start account state and return-law coefficients. On the bounded analysis domain of Assumption 3.1, the semi-synthetic return bound and transaction-cost rule keep each daily wealth multiplier uniformly positive and bounded. The fixed horizon therefore bounds the raw change in log wealth; reference values and holdings are bounded by the same assumption and the portfolio simplex. The finite factor-block library and compact parameter set keep Dirichlet shape parameters in a common compact subset of $(0, \infty)$. Uniform likelihood moments, the Lipschitz executable-action map in preceding holdings, and the fixed-horizon wealth and reference recursions give uniform sensitivity of the conditional outcome law to z and χ . The regularized CPT value and probability-weight functions then yield constants $C_1, C_2 < \infty$, independent of k and θ , such that

$$\sup_{\theta \in \Theta} |J_{k+1}(\theta, r_{k+1}) - J_k(\theta, r_k)| \leq C_1 (|\log X_{k+1} - \log X_k| + |r_{k+1} - r_k| + \|\omega_{k+1} - \omega_k\|_1) + C_2 \|\chi_{k+1} - \chi_k\|.$$

The episode-boundary transition gives $\|z_{k+1} - z_k\| = O(\alpha_k)$. The return-law coefficients change at one boundary, then along a finite interpolation, and are fixed thereafter; thus $\sum_k \|\chi_{k+1} - \chi_k\| < \infty$. Consequently $\mathcal{V}_K = O(K^{1/4}) = o(K)$ on the stated analysis domain. The training simulator samples the same conditional factor-block and return law as the environment, so its population gradient error is zero under that model.

E Proof of the Fixed-Dimension Minimax Rate

Proof of Theorem 5.14. We prove the lower and upper bounds under the same complete-trajectory oracle and the same expected trajectory budget. For the lower bound, it is enough to construct a two-point submodel contained in $\mathfrak{C}$ whose CPT gradients are separated by order $B^{-1/2}$ while the stopped transcript distributions remain at bounded total variation. For the upper bound, Theorem 5.6 supplies an unbiased finite-variance output with finite expected trajectory cost.

Step 1: a CPT-realizable one-stock submodel. Consider one risky asset and cash, horizon $h = 1$, zero transaction cost, and episode-start wealth and reference equal to one. Take the previous portfolio to be all cash and let the feasibility map be the identity. Set the risky feature to $e_1 \in \mathbb{R}^d$ and the cash feature to zero. At a query θ , the latent risky weight satisfies

$$U \sim \text{Beta}(e^{\theta^{[1]}}, 1),$$

where $\theta^{[1]}$ is the first coordinate of θ . After the inertia map (7), the executed risky weight is χU .

Let the risky return be $R = R_\star \xi$, where $R_\star > 0$ is fixed and sufficiently small and $\xi \sim \text{Bernoulli}(p)$ is independent of U . The terminal wealth and reference are

$$X_{\text{end}} = 1 + \chi R_\star U \xi, \quad r_{\text{end}} = 1 + \eta_+ \chi R_\star U \xi,$$

so the terminal relative outcome is

$$Y = v U \xi, \quad v = (1 - \eta_+) \chi R_\star > 0. \quad (88)$$

This is an admissible portfolio trajectory model under the common bounds in Assumption 3.6. The loss-side value is identically zero.

Step 2: local separation of the CPT gradient. Let $v = v_+$ and define

$$\mathbf{m}(t) = v v'(vt), \quad 0 \leq t \leq 1.$$

The shifted-power value function gives $\mathbf{m}(0) = 0$, $\mathbf{m}(t) > 0$ for $t \in (0, 1]$, and bounded $\mathbf{m}$. Since $P(U > t) = 1 - t^{e^{\theta^{[1]}}}$, a change of variables $y = v(vt)$ in the gain-side CPT integral gives

$$J_p(\theta) = \int_0^1 \mathbf{m}(t) w_+^\varepsilon \left(p \{1 - t^{e^{\theta^{[1]}}}\} \right) dt. \quad (89)$$

At the target $\theta_0 = 0$, differentiation under the integral is valid because, on a compact neighborhood of θ_0 , $t^{e^{\theta^{[1]}}} |\log t|$ has an integrable envelope on $(0, 1)$ and $(w_+^\varepsilon)'$ is bounded. Therefore

$$\nabla J_p(0) = \mathfrak{g}(p) e_1, \quad \mathfrak{g}(p) = p \int_0^1 \mathbf{m}(t) (w_+^\varepsilon)'(p(1-t)) (-t \log t) dt. \quad (90)$$

Put

$$A_v = \int_0^1 \mathbf{m}(t) (-t \log t) dt > 0, \quad d_W = (w_+^\varepsilon)'(0) > 0.$$

Since $|(w_+^\varepsilon)''| \leq C_2$, differentiation of (90) with respect to p gives

$$\begin{aligned} \mathfrak{g}'(p) &= \int_0^1 \mathbf{m}(t) (-t \log t) \left[(w_+^\varepsilon)'(p(1-t)) + p(1-t) (w_+^\varepsilon)''(p(1-t)) \right] dt \\ &\geq (d_W - 2C_2 p) A_v. \end{aligned}$$

Choose $p_{\max} \in (0, 1/2]$ small enough that $2C_2p_{\max} \leq d_W/2$; if $C_2 = 0$, any choice in $(0, 1/2]$ is admissible. Let

$$p_* = p_{\max}/2, \quad \mathcal{I} = [p_*/2, 3p_*/2], \quad c_* = d_W A_v/2.$$

Then

$$\mathbf{g}'(p) \geq c_* > 0 \quad \text{for every } p \in \mathcal{I}. \quad (91)$$

Step 3: information in an adaptive stopped transcript. For B large enough, take

$$p_{\pm} = p_* \pm \delta, \quad \delta = \delta_*/\sqrt{B},$$

where $\delta_* > 0$ is chosen so that $p_{\pm} \in \mathcal{I}$. For Bernoulli laws in the fixed interior interval $\mathcal{I}$, the elementary bound $\text{kl}(p\|q) \leq (p - q)^2 / \{q(1 - q)\}$ gives

$$\text{kl}(p_+ \| p_-) \leq C_I \delta^2 \quad (92)$$

for a finite constant C_I depending only on $\mathcal{I}$.

Consider an arbitrary admissible estimator in (65). Its query point at each oracle call may depend on the preceding transcript. Conditional on any such query, the latent Dirichlet action law is identical under p_+ and p_- ; only the Bernoulli return kernel differs. All algorithmic randomization can be included in the transcript and is likewise common to the two models. The data-dependent call count T_{call} is a stopping time for the oracle transcript filtration. The chain rule for the stopped likelihood ratio gives

$$\text{KL}(\mathbb{P}_+^{\text{tr}} \| \mathbb{P}_-^{\text{tr}}) = \mathbb{E}_+ T_{\text{call}} \text{kl}(p_+ \| p_-) \leq B C_I \delta^2 = C_I \delta_*^2. \quad (93)$$

The equality can be obtained by summing the conditional log-likelihood increments over $\{T_{\text{call}} \geq j\}$; these events are measurable before the j th fresh return is observed. The increments are uniformly integrable because $p_{\pm}$ stay in the compact interior interval $\mathcal{I}$ and $\mathbb{E}_+ T_{\text{call}} < \infty$.

Choose δ_* small enough that $C_I \delta_*^2 \leq 1/8$. Pinsker's inequality then gives

$$\text{TV}(\mathbb{P}_+^{\text{tr}}, \mathbb{P}_-^{\text{tr}}) \leq \frac{1}{4}. \quad (94)$$

Step 4: reduction from estimation to testing. By (91) and the mean-value theorem, the two gradient targets satisfy

$$D_B^g := \|\mathbf{g}(p_+)e_1 - \mathbf{g}(p_-)e_1\|_2 \geq 2c_*\delta. \quad (95)$$

Given any estimator $\hat{g}$, define the test that chooses the closer of the two targets. On a testing error, the estimator must be at least $D_B^g/2$ from the true target. The sum of the two testing error probabilities is at least one minus the total variation distance. Therefore

$$\begin{aligned} \max_{s \in \{+, -\}} \mathbb{E}_s \|\hat{g} - \mathbf{g}(p_s)e_1\|_2^2 &\geq \frac{(D_B^g)^2}{8} \{1 - \text{TV}(\mathbb{P}_+^{\text{tr}}, \mathbb{P}_-^{\text{tr}})\} \\ &\geq \frac{3c_*^2\delta_*^2}{8B}. \end{aligned}$$

The two models belong to $\mathfrak{C}$, so taking the infimum over all admissible procedures gives the lower bound in (66) with $c_{\text{lb}} = 3c_*^2\delta_*^2/8$.

Step 5: cost-matched upper bound. Fix any admissible estimator design parameters (n, ϱ, b) from [Theorem 5.6](#). The common model-class constants in [Assumption 3.6](#) make

$$\sup_{P \in \mathfrak{C}} \text{tr Cov}_P(Z_n^{\text{db}}) \leq C_{\text{var}}(n) < \infty, \quad C_{\text{cost}} = n(1 + \varrho \Upsilon_b) < \infty$$

uniform over $P \in \mathfrak{C}$. For

$$M = \left\lfloor \frac{\mathsf{B}}{C_{\text{cost}}} \right\rfloor,$$

run M independent, untruncated debiased outputs Z_n^{db} at θ_0 and average them. Their total expected number of complete trajectories is $MC_{\text{cost}} \leq \mathsf{B}$, so this is an admissible procedure. For $\mathsf{B} \geq 2C_{\text{cost}}$, $M \geq \mathsf{B}/(2C_{\text{cost}})$. Hence

$$\sup_{P \in \mathfrak{C}} \mathbb{E}_P \|\hat{g}_{M,n}^{\text{db}} - g_P\|_2^2 \leq \frac{C_{\text{var}}(n)}{M} \leq \frac{2C_{\text{var}}(n)C_{\text{cost}}}{\mathsf{B}}.$$

Thus the upper bound in [\(66\)](#) holds with $C_{\text{ub}} = 2C_{\text{var}}(n)C_{\text{cost}}$ and a sufficiently large B_0 . The lower and upper bounds use the same target, trajectory oracle, loss, and expected raw-trajectory budget, completing the proof. $\square$